Critical Design Evaluation



CRASH Rocket Project

Sponsor: UCF Senior Design

2/23/2026

Prepared by

John “Yianni” Bray (Aerospace)

Logan Hammer (Aerospace)

Camila Martinez (Aerospace)

Dylan Moga (Aerospace)

Christian Warson (Aerospace)

Faculty Advisor: Adam Marquardt

Table of Contents

1.0 Risk Assessment

- 1.1 Egg Payload
- 1.2 Water Payload Subassembly
- 1.3 Recovery & Propulsion
- 1.4 Aerostructures
- 1.5 Avionics

2.0 Modeling and Analysis

- 2.1 Egg Payload
- 2.2 Water Payload Subassembly
- 2.3 Recovery & Propulsion
- 2.4 Aerostructures
- 2.5 Avionics

3.0 Prototyping and Testing

- 3.1 Egg Payload
- 3.2 Water Payload Subassembly
- 3.3 Recovery & Propulsion
- 3.4 Aerostructures
- 3.5 Avionics

4.0 Development Planning

- 4.1 Egg Payload
- 4.2 Water Payload Subassembly
- 4.3 Recovery & Propulsion
- 4.4 Aerostructures
- 4.5 Avionics

1.0 Risk Assessment

1.1 Egg Payload

All potential failure modes of the egg payload system during the competition ultimately lead to the same negative outcome: loss of points in the competition. Documented risks could cause failure for the vehicle or a threat to safety, and can impact the team's score. The risks can be categorized into three groups: egg integrity, egg transfer, and capsule integrity. Egg integrity groups' failures that could damage the eggs, resulting in the team losing points for not protecting the eggs. Egg transfer group failures that could result in the team taking longer than allotted by the competition to transfer eggs, also resulting in losing points. Capsule integrity include potential failure points of the payload's structure, in simple terms, something breaking. All failure modes are confidently identified because the system was intentionally designed to be simple. With fewer factors in play, there are fewer failure points. The capsule is full of foam, a design that has been proven to work in past CRASH competitions and in egg drop competitions. Simple, reliable, and easy to find failure. The container for the capsule is a more experimental design to maximize the amount of foam and move the payload mass higher in the vehicle. So the container is under more scrutiny for risks.

The first category, egg's integrity, can be compromised by failure to dampen vibrations, failure to dampen impact forces, and failure to properly secure eggs. These risks can be negated by testing the payload system. This can be done by building a prototype of the capsule and dropping it from a significant height. Mimicking the typical egg-dropping competition. If it fails to protect the eggs, the team will see cracked eggs after a drop, easy to detect. Results will show the optimal design for the capsule's egg storage. The other category, egg transfer, can fail due to failure to unload within five minutes, the capsule being stuck in the rocket, the capsule not opening, the egg pucks taking too long to stack, or the egg pucks taking too long to unstack. This can be prevented by practicing loading and unloading the rocket before the competition. Practicing will reveal if the capsule has trouble opening or can get stuck, and adjustments to the capsule's shell can be made to prevent it in the future. The speed of loading and unloading can be refined by the team to be done in under 5 minutes. The categories of egg integrity and egg transfer do not affect the safety of attendants or risk the flight of the vehicle. These risks related to the egg payload system are manageable and preventable for the rocketry competition.

The third category, capsule integrity, includes the risk of the rod not being secured to the nose cone correctly, the snatchforce causing the rod to shear through the nose cone, and the snatch force shearing off the ubolt from the capsule. These would all result in the capsule being damaged, causing the vehicle to separate, resulting in the rocket not meeting NAR requirements for successful recovery. Disqualifying the team and risking the safety of the attendants of the competition with a free-falling object. This category is a priority to mitigate. The effects on the payload system by the snatch force will be evaluated with computer simulations. If the simulations are deemed insufficient, the team will default to a simpler design with more traditional methods of payload housing.

There are other risks not related to the competition, but rather to manufacturing. With the team being on a limited budget, there is a limited amount of material available for manufacturing. The most reliable way to mitigate the risk of eggs cracking during the flight is to test multiple configurations for the payload capsule. Testing the amount of eggs and their orientation. This requires multiple configurations to be manufactured, using a lot of material. The risk here is not having enough material for the final version of the capsule. To mitigate this, a CAD design will be created first, then the foam will be measured and marked before cutting. Reviews of video tutorials for creating foam capsules for regular egg drop competitions will also be done before manufacturing. This lowers the chances of mistakes being made during the manufacturing process, lowering the risk of not having sufficient material for the final product.

Table 1. Egg Payload Risk Assessment

Egg payload risks and prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Failure to protect the eggs	Failure to dampen vibrations	Can cause cracks	3	5	4	60	Testing to check that dampening is efficient
Failure to protect the eggs	Failure to dampen impact forces	Can cause cracks	3	5	4	60	Testing to check that dampening is efficient
Failure to protect the eggs	Failure to properly secure eggs	Can cause cracks	2	5	5	50	Test if egg pucks will rotate in the capsule or using two rods in the capsule
Failure to unload in under five minutes	Capsule stuck in the rocket	Can't load/ unload under 5 minutes	3	5	1	15	Test how the capsule fits in the rocket and practice loading and unloading
Failure to unload in under five minutes	Capsule not opening	Can't load/ unload under 5 minutes	1	5	1	5	Repeated practice of opening and closing the capsule
Failure to unload in under five minutes	Egg pucks take too long to stack	Can't load under 5 minutes	1	5	1	5	Practice stacking of pucks
Failure to unload in under	Egg pucks take too	Can't unload under 5	1	5	1	5	Practice the unstacking of

Egg payload risks and prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
five minutes	long to unstack	minutes					pucks
Failure to secure parts	Failure to secure the rod to the nosecone	The capsule and nose cone disconnect during flight	1	4	1	4	Practice the assembly of the payload system
Failure to secure parts	Failure of the nose cone to withstand the snatch force	The capsule and nose cone disconnect during flight	4	4	4	64	Simulate the structure under snatch forces
Failure to secure parts	Failure for bulkheads to withstand the snatch force	The payload disconnects from the vehicle during flight	2	5	4	40	Simulate the structure under snatch forces
Failure in manufacturing	Not enough Foam for the final design	Failure to deliver a reliable product	1	4	1	4	CAD of capsul, precise measurements before manufacturing, and prioritization of testing configurations.

1.2 Water Payload Subassembly

The water payload subassembly of the vehicle was analyzed using a failure modes and effects analysis to identify potential failure modes, their effects on the mission, and appropriate mitigations. The analysis covers all major subsystems of the subassembly, the HDPE vessels, the sealing system, the supporting structure, and a potential water deployment mechanism. It is important to note that any unintentional water leakage during the flight results in a disqualification from the competition, therefore all failure modes leading to water loss are treated with the highest severity. The FMEA scoring uses a 1-5 scaling for all three of the categories, likelihood, consequence, and detection, with a higher number denoting a worse performance for

the respective parameter. A general risk priority number was then calculated from these parameters to gauge the relative severity for each risk. The table below summarizes the findings from the FMEA. Each risk and the justification behind the RPN will be discussed, along with the mitigation strategies.

Table 2. Water Payload Risk Assessment

Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Vessel Structural Damage	Leakage resulting in disqualification	1	5	5	25	Ensure vessel can withstand forces with margin
Vessel Bursting	Leakage resulting in disqualification	1	5	5	25	Ensure vessel can withstand forces with margin
Vessel Thermal Damage	Structural integrity of the vessel compromised	1	3	4	12	Ensure vessel is shielded from direct thermal loads
Vessel Unplanned Detachment	Loose vessel which could become a projectile	2	4	5	40	Ensure vessel is properly seated in place
Caps Improperly Sealed	Leakage resulting in disqualification	2	5	3	30	Properly inspect sealing prior to launch
Caps Loosening due to Vibration	Leakage resulting in disqualification	2	5	5	50	Ensure caps are carefully and properly installed
Caps damaged due to Over-tightening	Leakage resulting in disqualification	2	5	4	40	Ensure caps are carefully and properly installed
Supports Structural Damage	Loose vessel which could become a projectile	1	4	5	20	Ensure supports can withstand forces with margin
Supports Unplanned Detachment	Loose vessel which could become a projectile	1	4	5	20	Ensure supports can withstand forces with margin
Supports Improperly Manufactured	Improper fittings exacerbating other issues	3	3	1	9	Ensure the supports are manufactured within specified margins

The HDPE bottles shall serve as the containment method for the water payload. A critical failure mode for the bottles is structural leakage, such as cracks. However the likelihood of the bottle cracking or suffering abrasion under normal conditions is very low, as the bottle material is strong [1], however there is the potential for manufacturing defects to induce cracks under stress from launch [2]. Obvious structural damage should be apparent upon basic inspection, however a manufacturing defect may not be detectable at all.

Another failure mode is the vessels bursting during specific high stress periods, such as takeoff, or impact on landing. The vessels being the failure point for such a case is extremely unlikely, but not impossible. As this failure would be sudden, detection is not possible. To mitigate such an event, any circumstances that could pressurize the bottles shall be avoided.

With the vessels in close proximity to the motor, a concern is the bottles experiencing thermal distortion from the heat generated by the motor burn. The material of the bottle was selected taking this into consideration [3], resulting in the likelihood of this failure mode being low, and designing the module such that the bottles are not in direct contact with the motor assembly shall be important to mitigate this risk. As with all modes related to the vessels, detection is not possible prior to flight.

A final failure mode for the vessels is becoming dislodged from their supporting structure. A loose bottle has the potential to become a projectile if it were to escape the rocket body. The vessel must stay removable from the vehicle, so complete attachment is not an option. It shall be important to firmly secure the vessels to the frame prior to takeoff. Detection is possible, through thoroughly inspecting the mounting of each vessel before the vehicle is launched.

Each vessel shall be sealed with a screw-on cap, with a secondary, redundant, sealing method. The cap and seal are critical to preventing leaks, and would be the most likely point of failure for the subassembly. The most preventable failure mode for this component is improperly installing the cap. A cap which is not fully threaded down will result in ineffective sealing, directly causing a leak [4]. The likelihood for this to occur is hopefully low, as with proper detection measures it will be caught before it is an issue.

A failure mode which could be exacerbated by the previously discussed mode, is the cap loosening due to vibrations during flight [5]. Due to the intensity of the forces experienced during flight, even a slight loosening of the sealing system could result in water leaking past the threads. Since the overall flight duration is not very long, the general likelihood of a vibration sustained for long enough to loosen the cap is low. Mitigation strategies for this mode are the same as before, with proper inspection and ensuring the sealing is properly fitted before flight.

A potential user error related failure mode could be induced by over-tightening the cap, damaging the neck of the bottle, or the cap itself. Such a crack might not be apparent until the bottle is subjected to intense flight conditions. The likelihood of this failure mode is low, as materials were selected with durability in mind. Mitigation is the same as before, with care being taken to ensure the sealing is properly fitted.

The support structure shall consist of rings which support the vessels around the motor. This structure's primary purpose is to secure the bottles in place during flight. A structural failure of this support system, such as a ring cracking or snapping, is a potential failure mode. If one of the rings were to break under the stress of flight, one or more vessels could come loose inside the body tube. Due to the location of the vessels, the consequence for this would not be very high, as they could not damage anything critical, however in the event they somehow escape from the vehicle during flight, the vessels could become a projectile. Therefore, robust construction of the lower module will ensure the vessels cannot escape the vehicle, if the rings were to crack.

A similar failure mode would be the mounting hardware dislodging completely during flight. Whether from vibration or other forces, if the structure completely becomes dislodged from the inner walls of the body tube, it could cause the motor to shift, which could be catastrophic for the flight trajectory of the vehicle. Mitigation of this failure mode will consist of using specific hardware to fasten the support structure to the body of the rocket.

Manufacturing errors have the potential to cause a misalignment or fitting issue which leads to a failure mode. If the cutouts are off-size, they could either be too tight or too loose for the water vessels. A tight fit could put additional stresses on the entire subassembly, while a loose fitting would allow the bottles to rattle. Either case has the potential to cause critical failures. The likelihood is moderate, as small discrepancies in measurements are not uncommon. Detection would be fairly simple, as a basic inspection of the components prior to assembly would reveal any potential misalignment errors. To mitigate this risk, a thin liner could be used to accommodate slight size variations, while ensuring the vessels fit snugly into place.

It is believed that this FMEA captures the entirety of the potential failure modes and mechanisms for the water payload subassembly. While every exact mode may not be explicitly stated, they would be similar enough in practice to be a derivation of one of the other modes already identified above.

1.3 Recovery & Propulsion

Practically every failure mode for recovery and propulsion will result in an invalid competition entry. This can range from our rocket not getting off the pad, to being recovered in an unflightworthy condition, to the complete destruction of our rocket. Due to the likelihood of any failures in these systems having detrimental effects on the outcome of our rocket, is it of the utmost importance to try to mitigate as many, if not all, failures as possible.

Out of the two systems covered in this section, the one that would have much worse results if a failure were to occur would be a recovery failure. If such a failure were to happen, it may result in the complete destruction of our rocket, which would require our team to completely rebuild, and in doing so, we may not be able to meet the deadline for our backup launch day. For example, if our main parachute failed to deploy fully, be it due to tangling in the lines or the material tearing, it would result in the vehicle not being able to fully slow down, which would lead to a ground collision at speeds the rocket was not designed for. This would destroy nearly every other system in the rocket and result in an invalid competition entry. On the other hand, a propulsion failure would be significantly less destructive, most likely keeping the rocket in a flightworthy condition. The highest likelihood failure would probably be a failed ignition, which while resulting in an invalid competition entry, would leave the rocket still in a flightworthy condition. The worst failure would probably be an ejection charge failure, but due to the main parachute deploying at its designated altitude, the rocket may be able to slow its descent down enough to result in minimal damage.

The severity of all failure modes is very high, but luckily nearly all of them have a low probability of occurrence. For example, the main parachute not deploying fully would be catastrophic, possibly resulting in the complete destruction of our rocket, but mitigating that failure would be rather simple by the team making sure the parachute is folded and packed correctly into the right bay, as well as making sure the shock cord is adequately sized to prevent any tangling. For propulsion, a failure of the motor resulting in an explosion would be very severe, leading to the destruction of our rocket, but the probability of occurrence is extremely

low since we will be using a motor manufactured by a reputable company with high quality control standards.

Detection of failures is very simple as nearly all of the failures in these systems result in clear visual issues. This however does not mean that it is easy to fix these problems on the fly. Due to the nature of a rocket, it is not possible to fix any issues that arise post launch, so our team cannot do anything about detectable failures while the rocket is in the air. We can however check things like the parachutes being packed correctly and making sure the motor is fully retained in the body.

To reduce possible failures, we can ensure that the motor, both parachutes, and the shock cord are free from visible damage, and get attached/loaded into the rocket correctly. The parachute must be folded correctly and loaded into the tube so that it can move freely inside, with the lines folded neatly so that they do not get tangled on deployment. The shock cord must be properly wrapped as well so it does not get tangled. And finally the motor must be properly inserted and retained so that it does not fall out. All actions to reduce possible failures are simple but will go a long way in ensuring our team's success.

All possible failure modes and mechanisms are believed to be identified as the FMEA below lists the most common failures in recovery and propulsion. If any failure modes and mechanisms have not been mentioned, they are likely of little to no importance and have a low severity and low possibility of occurrence.

Table 3. Recovery & Propulsion Risk Assessment

Recovery & Propulsion risks and prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Parachute Failure	Parachute tears	Rocket can't fully slow down	3	5	4	60	Ensure parachute is intact before launch
Parachute Failure	Parachute doesn't fully deploy	Rocket can't fully slow down	3	5	4	60	Ensure parachute lines aren't tangled before launch
Parachute Failure	Burned parachute	Rocket can't fully slow down	2	3	5	30	Ensure wad fully covers parachute
Parachute Failure	Parachute lines tangled	Rocket can't fully slow down	2	3	3	18	Ensure parachute lines aren't tangled before launch
Parachute Failure	Improper packing	Parachute may not	1	3	1	3	Ensure parachute is packed

Recovery & Propulsion risks and prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
		deploy					correctly
Parachute Failure	Improper folding	Parachute may not deploy correct	1	3	1	3	Ensure parachute is folded correctly
Parachute Failure	Wad plugging	Parachute may not deploy	1	5	2	10	Ensure wad can move freely in body tube
Parachute Failure	Broken parachute lines	Possible parachute loss	2	4	1	8	Ensure parachute lines are intact before launch
Parachute Failure	Inadequate parachute size	Rocket can't fully slow down	2	5	4	40	Test with simulators to attain correct parachute size
Parachute Failure	Early deployment	Rocket drifts too far	4	5	5	100	Ensure altimeter is set correctly
Parachute Failure	Late deployment	Rocket can't fully slow down	4	5	5	100	Ensure altimeter is set correctly
Shock Cord Failure	Shock cord snaps	Parachute loss	2	5	1	10	Ensure shock cord isn't damaged before launch
Shock Cord Failure	Zippering	Body tube damage	3	1	1	3	Ensure zipper protectors are placed correctly
Shock Cord Failure	Shock cord gets tangled	Shock cord doesn't reach full length	2	2	3	12	Ensure shock cord is folded correctly
Shock Cord Failure	Shock cord gets burned	Possible parachute loss	1	4	3	12	Ensure wad is low over motor
Motor Failure	Motor explodes	Invalid competition entry	1	5	1	5	N/A

Recovery & Propulsion risks and prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Motor Failure	Ejection charge fails	Drogue does not deploy	2	4	1	8	Backup ejection charge for drogue
Motor Failure	Motor case fails	Rocket may explode	1	4	5	20	N/A
Motor Failure	Motor fails to ignite	Rocket does not launch	1	5	1	5	Ensure igniter isn't damaged
Motor Failure	Motor falls out	Rocket loses propulsion	1	5	5	25	Ensure retaining spring and centering rings are undamaged
Motor Failure	Motor wall burn-through	Heat may burn through rocket tube	1	3	5	15	N/A
Motor Failure	Nozzle blow-out	Motor loses thrust	1	5	5	25	N/A
Motor Failure	Premature ejection	Target altitude not reached	1	5	5	25	N/A

1.4 Aerostructures

Failure conditions for the aerostructures system of a high powered rocket can ultimately lead to total rocket destruction and lead to immense safety concerns. This body system must balance the structural properties, payload security, and the propulsion system. A previously compiled FMEA stated that multiple failure modes could lead to total mission failure and or potentially hazardous conditions. Main concerns for the aerostructure include buckling of the airframe under intense loads or compromisation of the body tube integrity, bending or chipping from the fins or nosecone leading to an improper flightworthy condition, fluttering of the fins leading to deviation from planned trajectory and other aerodynamic parameters, and failure of fastening components causing the rocket to break up during launch. Additionally, there is the potential for material fatigue and cracking to occur, potentially compromising the entire system and leading to failure.

These failures are considered high risk and therefore must be mitigated at all costs. Attention to detail when it comes to quality control and ideal overall design to maintain proper

flight worthiness. Due to this flight being short, some of these risks can be considered moderate or low probability, but most, if not all are critical to proper completion of the competition.

Detecting these defects can often be noticeable prior to launch. Analyzing the overall structure for proper stability such as the body being clear of buckles, cracks, and other distortion will allow the rocket to proceed to launch. Fin and nose integrity is critical to launch performances and shall be clear of deformation. As for when the rocket is set for launch and on the pad, the ability to make alterations is limited. Only for crucial safety hazards would the team be able to alter the rocket when safe to do so. Throughout inspection of all materials will have to be complete to ensure all components are securely fastened and the body is free of deformation. Internal defects and stresses will also be difficult to detect because the deformations will be less visible and would have to be analyzed through destructive testing.

To avoid these failures, the aerostructure has been critically designed to account for weight distribution, stability, and correct positioning of the different payloads and subsystems. Careful material selection has been done to ensure the stability of the rocket components through prior experimentation and previous research. Proper quality control will be necessary when manufacturing each component and fabricating the entire rocket body itself. Pre-flight inspection will be crucial to approving the vehicle for launch as it shall be free of defects critical to performance. Using the team's FMEA and prior rocket design and launching knowledge, the team has made sound judgement of the materials selection and overall design for a safe and successful launch. Validation of these systems and inspections shall reduce the likelihood of failed critical materials and systems.

Table 4. Aerostructures Risk Assessment

Aerostructures Risks & Prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Body Tube Failure	Body tube cracks	Could prevent reuse or reduce stability	2	3	2	12	Handling the rocket carefully and controlling descent velocity
Body Tube Failure	Body tube get burnt through	Cause a hole leading to decreased stability and prevent reuse	2	3	1	6	Containing motor burn or carefully positioning motor to avoid burning
Body Tube Failure	Body tube disconnects early	Invalid competition entry as altitude will not be reached	1	5	1	5	Ensure proper connection

Aerostructures Risks & Prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Body Tube Failure	Body tube doesn't separate	Safety violation as rocket will have no decreased velocity	1	5	1	5	Ensure proper connection
Body Tube Failure	Body tube fails to bond with centering rings	Motor tube disconnects leading to safety hazard	1	5	3	15	Ensure proper connection & build up epoxy
Body Tube Failure	Body tube fails to properly bond with nose cone	Rocket becomes unstable and crashes	1	5	3	15	Ensure proper connection & build up epoxy
Body Tube Failure	Rail pins create crack propagation	Could prevent reuse depending on crack size	3	3	2	18	Carefully screw into rocket
Body Tube Failure	Rail pins tear through body	Tear could prevent reuse	1	3	1	3	Ensure proper connection and screw is not fretting
Body Tube Failure	Body tube buckles	Instability; rocket crashes	1	5	1	5	Carefully handle rocket and avoid coming in contact with liquid
Nose Cone Failure	Nose cone disconnects during flight	Instability; rocket crashes	1	5	1	5	Ensure proper connection
Nose Cone Failure	Nose cone buckles during flight	Instability; rocket crashes	1	5	2	10	Ensure material is stable & has a strong bond
Nose Cone Failure	Nose cone cracks	Could prevent reuse or reduce stability	2	3	2	12	Handling the rocket carefully and controlling descent velocity

Aerostructures Risks & Prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Nose Cone Failure	Nose cone dents	Could prevent reuse or reduce stability	3	3	1	9	Handling the rocket carefully and controlling descent velocity
Fin Failure	Fins crack before/during ignition	Could lead to additional cracking upon landing; prevent reuse	4	2	2	16	Handling the rocket carefully and controlling descent velocity
Fin Failure	Fins chip before/during ignition	Could affect flight performance/stability	2	4	1	8	Handling the rocket carefully and controlling descent velocity
Fin Failure	Fins break off before/during ignition	Extreme hazard/safety concern	1	5	1	5	Handling the rocket carefully and controlling descent velocity
Fin Failure	Fins break off during landing	Prevents reuse entirely; defies competition rules	2	5	1	10	Handling the rocket carefully and controlling descent velocity
Fin Failure	Fins crack during landing	Compromises fin integrity	3	1	2	6	Handling the rocket carefully and controlling descent velocity
Fin Failure	Fins chip during landing	Potentially prevents reuse depending on size	4	4	1	16	Handling the rocket carefully and controlling descent velocity
Fin Failure	Fins fail to bond to the body tube	Fins fall off, leading to instability	1	5	1	5	Ensure proper connection & build up epoxy
Fin Failure	Fin material shears	Compromises fin integrity	1	2	2	4	Delicate Manufacturing & material handling
Fin Failure	Fins burn during ignition	Compromises fin integrity	3	3	1	9	Containing motor burn or carefully positioning fins to

Aerostructures Risks & Prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
							avoid burning
Fin Failure	Fins misaligned for launch	Could lead to instability	2	5	3	30	Ensure proper alignment, connection, and epoxy build-up
Fin Failure	Uneven fin surfaces	Compromises fin integrity and air flow	1	2	3	6	Delicate Manufacturing & material handling
Fin Failure	Fins flutter during flight	Causes instability	2	5	2	20	Ensure proper alignment, connection, and epoxy build-up
Fin Failure	Fins bend during flight	Causes instability	2	5	3	30	Ensure material is stable & has a strong bond
Bulkhead Failure	Bulkhead fails to bond with body tube	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
Bulkhead Failure	Bulkhead fails to bond with coupler	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
Bulkhead Failure	Bulkhead fails to bond with payload bay(s)	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
Bulkhead Failure	Bulkhead cracks	Compromises bulkhead integrity	2	1	2	4	Handling the rocket carefully and controlling descent velocity
Bulkhead Failure	Bulkhead chips	Compromises bulkhead integrity	2	4	1	8	Handling the rocket carefully and controlling descent velocity
Bulkhead Failure	Bulkhead disconnects/bursts	Rocket integrity compromised	2	5	1	10	Ensure proper alignment, connection, and

Aerostructures Risks & Prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
		causing launch failure					epoxy build-up
Bulkhead Failure	Fretting observed between eye bolt & bulkhead	Possible disconnect of eye bolt and therefore recovery system failure	2	5	2	20	Delicate Manufacturing & material handling
Bulkhead Failure	Eye bolt tears out of bulkhead	Recovery system failure	2	5	1	10	Ensure material is stable & has a strong bond
Bulkhead Failure	Bulkhead burns	Compromises bulkhead integrity	1	2	1	2	Containing motor burn/boom pool
Centering Ring Failure	Centering ring(s) fail to bond with body tube	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
Centering Ring Failure	Centering ring(s) fail to bond with motor tube	Disconnects leading to unflightworthy state	2	5	4	40	Ensure proper connection & build up epoxy
Centering Ring Failure	Centering ring(s) crack	Compromises centering ring integrity	2	1	2	4	Handling the rocket carefully and controlling descent velocity
Centering Ring Failure	Centering ring(s) chip	Compromises centering ring integrity	2	4	1	8	Handling the rocket carefully and controlling descent velocity
Centering Ring Failure	Centering ring(s) disconnect/burst	Rocket integrity compromised causing launch failure	2	5	1	10	Handling the rocket carefully and controlling descent velocity
Centering Ring Failure	Fretting observed between eye	Possible disconnect of eye bolt and	2	5	2	20	Ensure proper alignment, connection, and

Aerostructures Risks & Prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
	bolt & centering ring	therefore recovery system failure					epoxy build-up
Centering Ring Failure	Eye bolt tears out of Centering ring	Recovery system failure	2	5	1	10	Ensure material is stable & has a strong bond
Centering Ring Failure	Centering ring(s) burn	Compromises centering ring integrity	1	2	1	2	Containing motor burn/boom pool
Motor Retainer Failure	Female side of retainer disconnects from bottom centering ring	Lack of motor security; Possible motor loss/instability	2	5	1	10	Ensure proper connection & build up epoxy
Motor Retainer Failure	Male side of retainer unscrews during launch	Lack of motor security; Possible motor loss/instability	1	5	1	5	Ensure proper connection
Motor Retainer Failure	Retainer cracks	Integrity compromised	2	2	2	8	Handling the rocket carefully and controlling descent velocity
Motor Retainer Failure	Retainer chips	Integrity compromised	1	3	1	3	Handling the rocket carefully and controlling descent velocity
Motor Retainer Failure	Retainer burns and/or melts	Shape failure and possibly unable to hold motor	2	4	1	8	Containing motor burn
Motor Retainer Failure	Retainer bursts	Rocket integrity compromised causing launch failure	2	5	1	10	Handling the rocket carefully and controlling descent velocity

1.5 Avionics

The avionics subsystem has been assessed for failure modes that would result in the system falling short of its projected goals in the project. Each failure has been categorized in types such as structural, power, electrical, mechanical and environmental. Under each category, there are various modes of failure that vary in effect on the vehicle's function. As the avionics system holds both basic structural and complex electrical components, it has a large variety of potential failures.

In order to evaluate the severity of each effect, each potential failure mode was listed with its effect on the overall system, likelihood of its occurrence, consequence of the failure, and how easily it can be detected. The last three of those aspects were quantified on a scale of ten and then multiplied together to give them a Risk Priority Number (RPN), which allows the responsible engineer to gauge the relative importance of failures compared to others.

Based on these scores, some of the highest priority failure modes were determined to be U-bolt shearout, black powder ignition failure, screw-terminal wire deseating, and premature battery depletion. These items are all then given mitigation procedures or precautions that can be incorporated in the building or operation of the vehicle. Below is the compiled list of potential failures, effects, RPNs and mitigation techniques.

Table 5. Avionics Risk Assessment

Avionics Risks and Prioritization

Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Power Failure	Premature battery depletion	Complete loss of FC and camera functions	2	5	3	30	Size battery (mAh rating) for long pad operations
Electronics Failure	Flight controller dropout (malfunction)	Loss of reliable flight event control and data logging	1	5	2	10	Ensure proper handling and operation procedure is followed
Mechanical Failure	Screw terminal wire de-seating	Loss of power to electrical functions mid-flight	3	5	2	30	Secure all wires with items like zipties
Electronics Failure	Camera dropout (malfunction)	Loss of recorded video (bonus points)	2	2	2	8	Ensure large enough SSD storage and operation

Avionics Risks and Prioritization							
Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
Environmental Failure	Camera fogging obstruction	Loss of video quality	1	1	4	4	Minimize temperature change and visually inspect
Environmental Failure	Electrical component condensation failure (Humidity)	Potential loss of reliable flight event control and data logging	1	5	4	20	Minimize temperature change to prevent condensation
Environmental Failure	Electrical component overheat failure (Temperature)	Potential loss of reliable flight event control and data logging	1	5	4	20	Ensure proper component venting and spacing
Mechanical/Environmental Failure	Black powder ignition failure	Failure of primary recovery deployment	2	5	4	40	Keep powder dry and securely packed
Electronics Failure	Audio recovery device failure	Loss of ground recovery aide	3	2	2	12	Only enable device as close to before launch as possible
Electronics Failure	Unintentional arming switch toggle	Potential premature recovery event	1	4	1	4	Ensure arming switch is secured in either operating position
Mechanical Failure	Avionics bay integrity failure on ground hit	High potential for electronics damage	2	4	2	16	Minimize ground-hit velocity
Mechanical Failure	Avionics bay integrity failure on recovery deployment	High potential for electronics damage	2	4	2	16	Minimize ground-hit velocity
Mechanical Failure	U-Bolt shear out	Recovery failure; potential freefall	3	5	3	45	Use thick bulkheads and separate at apogee
Mechanical Failure	E-sled improperly	Probable component	2	3	3	18	Ensure all parts are within proper

Avionics Risks and Prioritization

Type of Failure	Failure Modes	Effects	Likelihood (L)	Consequence (C)	Detection (D)	RPN (LxCxD)	Mitigation
	secured	deseating or additional vibration stress					tolerance and torque
Mechanical Failure	Components improperly secured to e-sled	Probable component deseating or additional vibration stress	2	3	3	18	Ensure all parts are within proper tolerance and torque

2.0 Modeling and Analysis

2.1 Egg Payload

The system concept is visualized using SolidWorks computer-aided design. Using CAD is useful for better visualizing the size of the eggs in relation to the diameter of the rocket. Depicting what can reasonably fit in the capsule and the amount of foam available for cushioning. The model will be accurate for the cardboard, foam, rod, and bolts, but will likely vary in the cutouts, depending on the variation in egg sizes. The CAD will be based on the average size of eggs in the United States, but will be prototyped using real store-bought eggs. The parameters of the egg capsule are the limits imposed by the vehicle, such as size and weight. The capsule must fit within the vehicle, so it must have a diameter under 4in, and the weight is limited to 32oz .To calculate cushioning, a website was created to find the optimal foam material and distance for cushioning. The method for calculating optimum foam density and distance between eggs is derived from the MIT Structures lesson Cellular Solids: Structure, Properties and Applications [6].

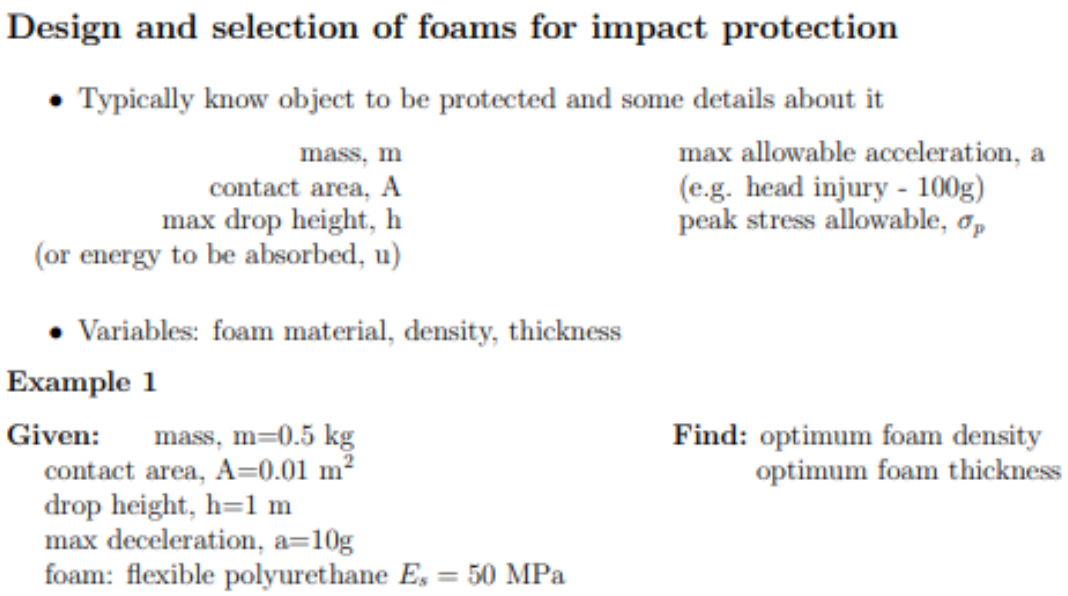


Fig 1. Slide from MIT Structures lesson Cellular Solids: Structure, Properties and Applications [6]

Example 1:
Find Foam Density and Thickness

Table 8.2 Example 1: selection of foams

Specification of the problem		
Mass of the package object, $m = 0.5\,\text{kg}$		
Area of contact between foam and object, $A = 0.01\,\text{m}^2$		
Velocity of package on impact (drop height $h = 1\,\text{m}$), $v = 4.5\,\text{m/s}$		
Energy to be absorbed, $U = mv^2/2 = 5\,\text{J}$		
Maximum allowable package force (based on deceleration of 10g), $F = ma = 50\,\text{N}$		
Maximum allowable peak stress, $\sigma_p = F/A = 5\,\text{kN/m}^2$		
Solid modulus in foam (flexible polyurethane), $E_s = 50\,\text{MN/m}^2$		
Maximum allowable normalized peak stress, $\sigma_p/E_s = 10^{-4}$		
Iterative procedure		
1st Iteration	$t_1 \gg 1$	$t_1 \ll 1$
Initial choice of t_1	1 m	0.001 m
Resulting strain-rate, $\dot{\varepsilon} = v/t_1$	$4.5\,\text{s}^{-1}$	$4.5 \times 10^3\,\text{s}^{-1}$
Resulting (W/E_s) at $\sigma_p/E_s = 10^{-4}$	5.25×10^{-5}	7.4×10^{-5}
Energy absorbed per unit volume, W	$2620\,\text{J/m}^3$	$3700\,\text{J/m}^3$
2nd Iteration		
Revised t_2 (from $U = WAt$)	0.19 m	0.14 m
Revised $\dot{\varepsilon} = v/t_2$	$24\,\text{s}^{-1}$	$32\,\text{s}^{-1}$
Revised (W/E_s)	6.6×10^{-5}	6.7×10^{-5}
Revised W	$3300\,\text{J/m}^3$	$3350\,\text{J/m}^3$
3rd Iteration		
Revised t_3 (from $U = WAt$)	0.15 m	0.15 m
Optimum density, ρ^*/ρ_s (Fig. 8.8)	A little below 0.01	

Fig 2. Slide from MIT Structures lesson Cellular Solids: Structure, Properties and Applications [6]

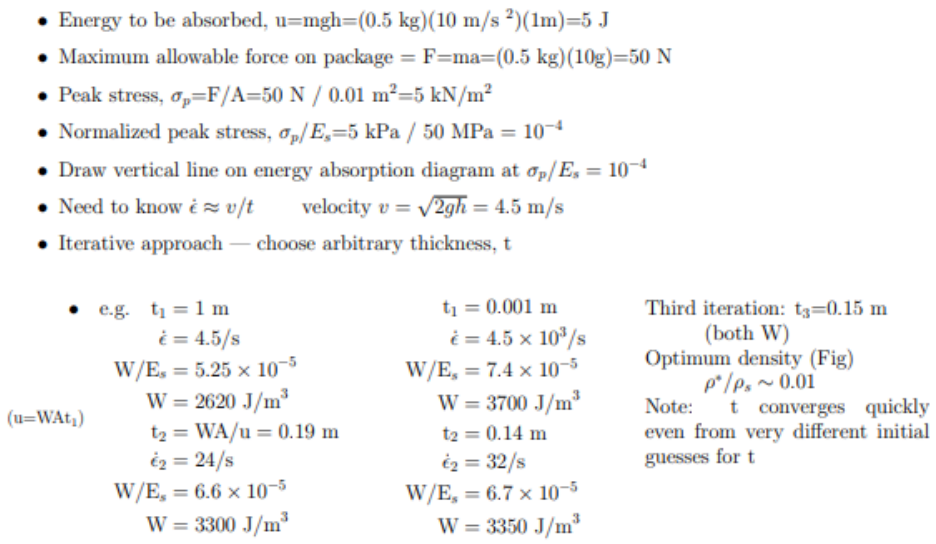


Fig 3. Slide from MIT Structures lesson Cellular Solids: Structure, Properties and Applications [6]

The iterative process described in the example was translated into a computational code developed for this project to automate and streamline the evaluation of foam materials. This code also allows the ability to choose between charts for Polyethylene Foam Sheet (Closed-Cell PE) or Cross-linked Polyethylene Foam (XLPE), data was collected from Foam Factory's foams' physical data sheets [7]. Young's modulus values for the foam were approximated from the manufacturer's compression-deflection data (ASTM D3575) by computing an effective secant modulus in the 25–50% strain range. A Poisson's ratio of 0.25–0.30 was assumed, consistent with reported values for conventional closed-cell polyethylene foams.

Table 6. Comparison of PE and XPLE Foams

Name of foam	Density (lb/ft³)	Compressive Strength (lb/in²) at 25%	Compressive Strength (lb/in²) at 50%	Young's Modulus, E (psi)	Poisson's Ratio, v
Polyethylene Foam Sheet (Closed-Cell PE)	1.2	7	12	~26	0.25
Polyethylene Foam Sheet (Closed-Cell PE)	1.7	7	14	~28	0.25
Polyethylene Foam Sheet (Closed-Cell PE)	2.2	8	14.5	~31	0.25
Polyethylene Foam Sheet (Closed-Cell PE)	4	16.5	20	~53	0.25
Cross-linked Polyethylene Foam (XLPE)	2	8.2	17.1	~34	0.25–0.30
Cross-linked Polyethylene Foam (XLPE)	3	10.5	19.8	~41	0.25–0.30
Cross-linked Polyethylene Foam (XLPE)	4	21.2	31.8	~74	0.25–0.30

Link to webpage: https://camilaarmartinez.github.io/foam-impact-tool/



Fig 4. Foam Impact Design Calculator

Increasing the allowed force on the eggs reduces the distance of the foam needed. If restricted to a limit of 35N acting on the egg, a distance of above 10cm for PE and 4cm for XPLE. A limit raised to 100N requires around 5cm for PE and 1.5 cm for XPLE. A limit raised to 150N requires around 3.5cm for PE and 1 cm for XPLE. A limit raised to 200N requires around 2.7 cm for PE and 0.7 cm for XPLE. The study by ADMET [9] tested eggs using a single-column compression machine with foam padding and found that eggs broke under an average load of 53 lb (235.8 N) on their poles and 90 lb (400.3 N) on their sides. This is a much

higher force than the study cited for M2 and the tech memos. This study is more relevant to the project, focusing on the distribution of force on the egg instead of the egg's resistance to dropping. A padding of 10cm PE for a maximum force of 35N is impractical and not space-efficient. With a body tube diameter of 4 inches, the maximum space between eggs and the wall is 3.13cm.

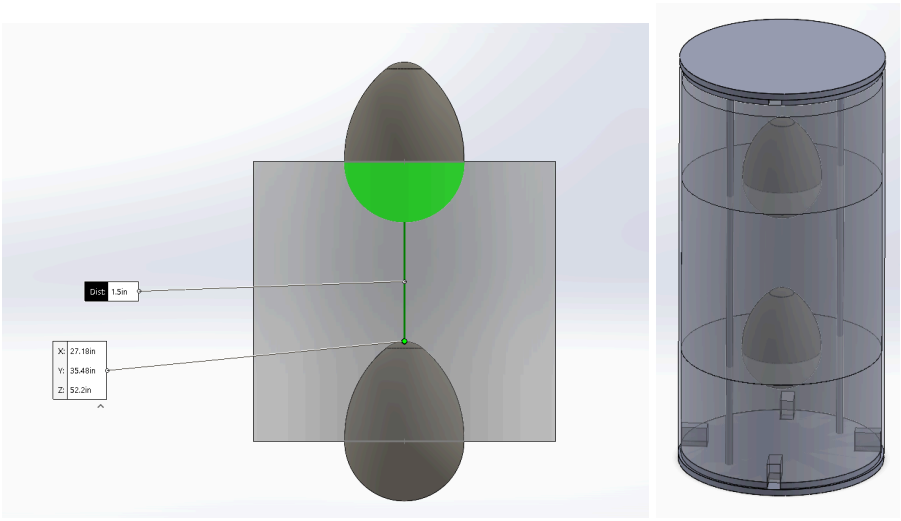


Fig 5. 3.5 in PE 2.2

PE 2.2 would be slightly below the optimal 3.4cm for 150N, having 3.13cm of foam between the eggs and the wall. With the option of cutting a premade cylinder instead of sheets, the height of the pucks can be adjusted to the optimal 3.4cm of spacing with a height of 3.5 inches. This PE cylinder foam will cost $14.99. With a max height of 8 inches for the sleeve of the capsule, there would only be 2 eggs carried. When disregarding cushioning optimization, the design can instead be optimized for maximum egg capacity within the capsule. For each additional egg over two that is safely recovered, the team earns 10 additional points. So, for point maximization, potentially having more eggs, assuming some but not all break, will earn more points than only launching two.

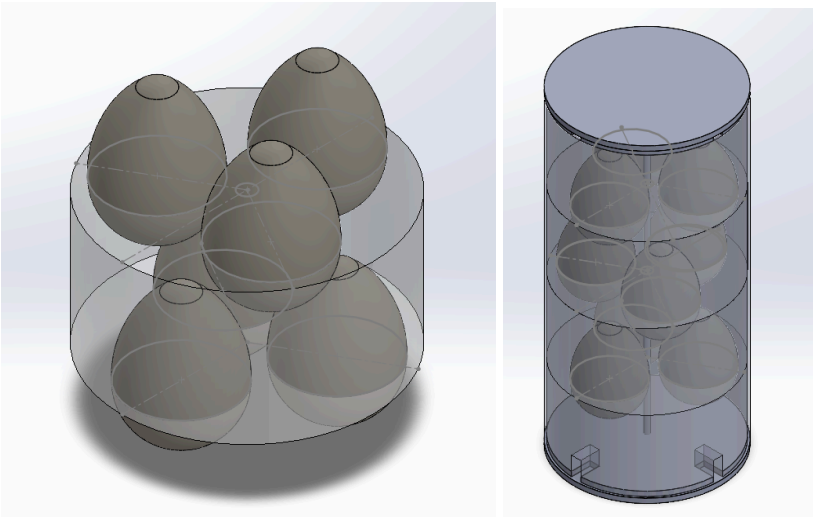


Fig 6. 2in PE 2.2 EGG 3

Maximized to have 3 eggs per layer in 2-inch pucks, 9 eggs total fitting in the coupler.

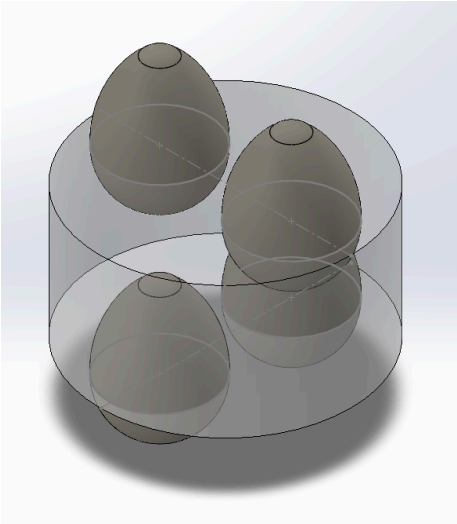


Fig 7. 2in PE 2.2 EGG 2

Maximized to have 2 eggs per layer in 2-inch pucks, 6 eggs total fitting in the coupler.

Table 7. Egg Capsule variation measurements

Variation	Egg capacity	Cost	Weight (foam + eggs)
XPLE 2 2 in	3	$57.99	7.76 oz
XPLE 2 3in	2	$77.99	5.79 oz
PE 2.2 3.5in	2	$14.99	5.97 oz
PE 2.2 2in EGG 3	9	$14.99	19.69 oz
PE 2.2 2in EGG 2	6	$14.99	13.83 oz
Dual egg protector from Aerotech	2	$11.50	5.94 oz

Additional weight for the Egg payload includes the coupler, aluminum centering rod, U-bolts, and 2 plywood bulkheads.
Coupler: 6.38 oz
Aluminum rod (¼" × 8"): 0.61 oz
Two U-bolts: 2.04 oz
Two plywood bulkheads: 2.32 oz
Totaling to 11.35 oz.

Table 8. Egg Capsule Decision Matrix

Variation	Egg capacity (1-5)	Low Chance of breaking (1-5)	Cost (1-5)	Weight (1-5)	Manufacturing (1-5)	Total
XPLE 2 2 in	1	4	1	4	2	17
XPLE 2 3in	1	5	1	5	2	20
PE 2.2 3.5in	1	5	5	5	5	25.5
PE 2.2 2in EGG 3	5	1	5	1	5	25.5
PE 2.2 2in EGG 2	4	1	5	2	5	23.5
Dual egg protector from Aerotech	1	5	3	5	5	23.5

The egg capacity is weighed times 3, the chances of breaking are weighed times 2, and manufacturing is divided in half. PE with egg capacity of 2 eggs, the competition minimum, and the PE with egg capacity of 9, have the same score in the decision matrix. The version of maximum eggs would weigh 31.04 oz, and the maximum cushioning would weigh 17.32 oz. Because both options use the same material and require purchasing a minimum 17-inch cylinder, both versions can be manufactured and tested before launch at no extra cost. This allows the team to choose the final launch version based on test results.

These CADs of the foam internals inform the variation of prototypes that will be manufactured and tested. With the ability for the team to decide to use the configuration with the higher risk but greater reward of points, or use the safer configuration for guaranteed points. While the calculation for egg safety predicts that only 2 eggs could survive, it has been proven by past teams that eggs are very resilient, especially with a soft landing. Despite the calculations leaning to a conservative conclusion, prototyping and testing will yield the most accurate results.

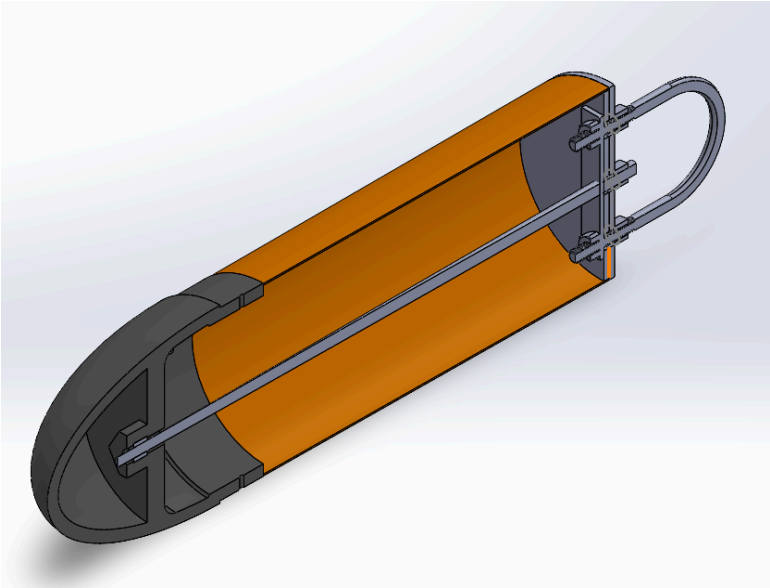


Fig 8. Nose cone integrated payload

The experimental nose cone integrated version of the capsule was designed separately, as the externals of the capsule can be changed based on risk assessment through simulation. A traditional payload design would have the nose cone attached to the payload with a shockcord; the experimental design integrates the two components. Allowing the team to use space in the nosecone, theoretically, making the egg transfer process more streamlined, increases foam per egg, and moves mass upwards. The payload's rod will go through the nose cone and thread into a nut placed during the 3D printing of the cone. The amount of material between the nut and the surface toward the rod increases the amount of force that can be loaded on the cone. This will be simulated with ANSYS to determine if the 3D printed cone will not fracture under the load of the snatch force. The payload will be secured to the rocket by screws through the plastic nose cone. The other end of the payload will have a ubolt secured to the bulkheads for the shock cord to attach to.

2.2 Water Payload Subassembly

The complete water payload subassembly has been modelled using SOLIDWORKS. This 3D CAD model is what currently serves as the team's primary method of visualizing the system.

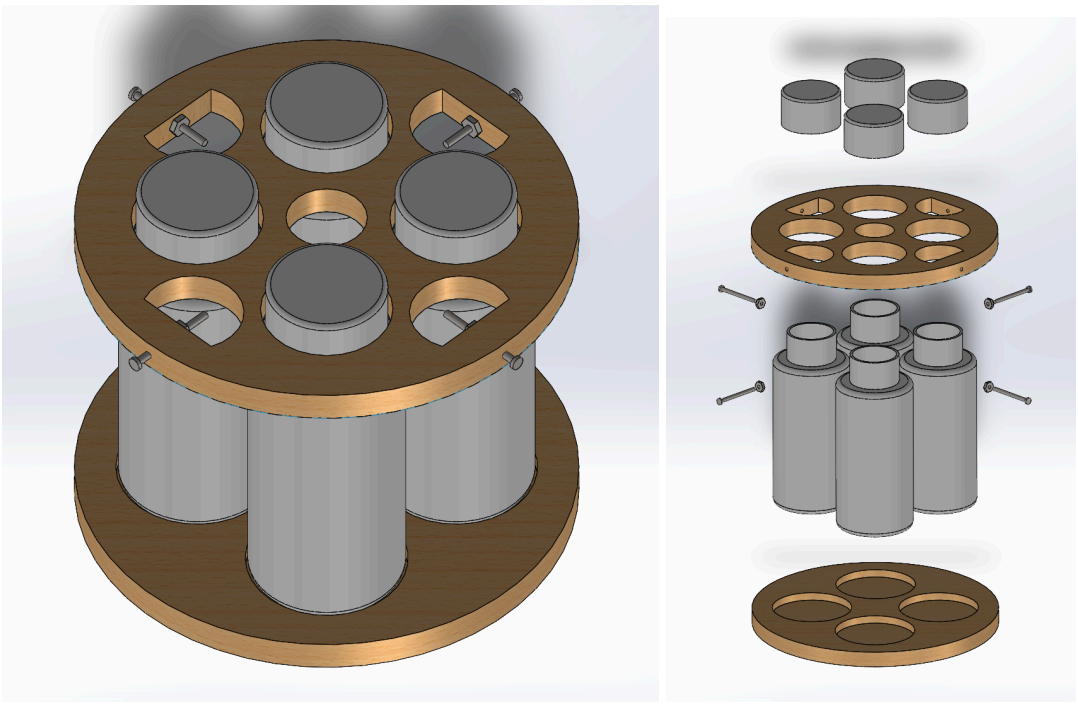


Fig 9. Water Payload Subassembly Models

The main engineering principles applied to this system are Newton's second law, hydrostatics, stress, and vibrations. Using these principles, an analysis of the water payload was performed to gather a baseline for the forces which must be sustained by the subassembly during flight. This information is important as under-designing could result in disqualification from the

competition due to water spillage. The primary objective of this analysis is to ensure that the water vessels, the seals, and the supporting structure are able to withstand the forces sustained during peak acceleration. To proceed with the analysis, several assumptions about the design shall be defined.

Newton's second law [10] was used to calculate the total force to be experienced by the supports due to the water payload. Based on prior simulations of the proposed launch vehicle, the max acceleration to be experienced is 173.4 m/s^2, or about 18g. The combined mass of each complete vessel and its payload is estimated as 0.0552kg, or 1.95oz. Each individual module will produce a force equal to the peak acceleration multiplied by its mass.

$$173.4\left(\frac{m}{s^2}\right) \times 0.0552\ kg \approx 9.6\ N$$

Therefore with four vessels a total force of 38.4 N, or about 8.6 lbf, is the total axial load that the support system has to carry during launch.

Another force which could test the structural integrity of the assembly is the pressure experienced by the caps of the vessels. The density of water is 1000 kg/m^3 [11], and the height of the vessel is 3 inches, or 0.0762 m. With these, the simplified Stevin's Law [12] was used to formulate the hydrostatic pressure on the caps during peak acceleration.

$$1000\left(\frac{kg}{m^3}\right) \times 173.4\left(\frac{m}{s^2}\right) \times 0.0762\ m \approx 13.2\ kPa$$

This combined with the pressure force due to the mass of the water over the area of the cap will give the max pressure on the cap. The vessels will have a standard 20-410 neck fitting, giving an inner cap diameter of 20mm. The mass of the water, multiplied by max acceleration, divided over the area of the cap shall give the pressure force.

$$0.0354369\ kg \times 173.4\left(\frac{m}{s^2}\right) \div \frac{\pi}{4}(0.02\ m)^2 \approx 19.6 kPa$$

The results from both of these equations is a max pressure on the cap of about 32.8 kPa, or about 4.8 psi.

The results of these basic preliminary calculations confirm that the subassembly can be designed to withstand the peak launch loads with significant margin. Findings showed that the overall pressure loads are minor, and the total force induced by the payload is not much more significant.

2.3 Recovery & Propulsion

Unlike the other systems outlined in this section, recovery and propulsion are much simpler as we are using the same methods that the majority of other model rockets use. This method is very well defined as it has decades of successful use.

The team will visualize the system with Figure 10 below. The ejection charge from the motor will eject the drogue chute to reduce descent velocity without allowing the rocket to drift and then the second ejection charge will eject the main chute to reduce descent velocity in the final seconds of descent.

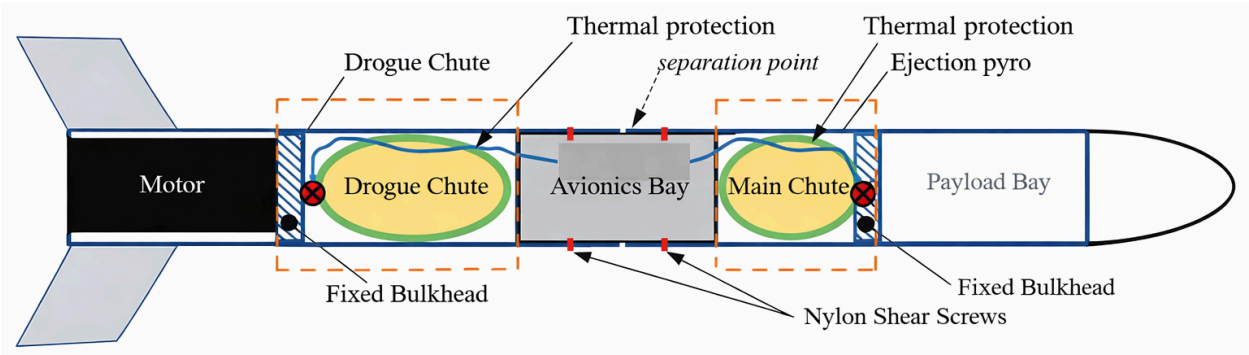


Fig 10. Model of recovery system in a dual deployment rocket [13]

The fundamental engineering principles applicable to these systems are conservation of momentum, thermodynamics, and aerodynamics. With the conservation of momentum we can simulate the altitude that our rocket will reach. With thermodynamics we can find temperatures of things like the motor and ejection charges and ensure that no components receive higher temperatures than what they are designed for. With aerodynamics we can calculate descent velocities with the parachute sizes we have selected as well as how far our rocket may drift due to wind.

Modeling and analysis will be done through the use of OpenRocket simulations. We have run simulations with different motor types as well as different parachute sizes in order to guarantee that our rocket meets all project requirements. Analysis is done via analyzing graphs and charts created by the software with the information obtained through the simulations.

Relevant parameters are total impulse, thrust curve, mass, drag, center of pressure, and center of gravity. The total impulse of the motor is 320 Newton seconds; mass is undetermined since we have testing to do on other components, which means drag is also undetermined; and values for center of pressure and center of gravity aren't as important, but making sure the center of gravity is above the center of pressure is.

A modeling and analytical approach is an accurate representation of the physical system because programs like OpenRocket have been used for years and are known to be reliable and provide accurate analysis of different rockets. The program compiles centuries of mathematical and physical knowledge in order to create simulations with values within 5-10% of what would occur in a real-life scenario.

The boundary conditions are things such as the mass of the motor and the thrust that it provides, and the power of the ejection charges. The constraints are things such as the motor impulse, stability, wind, and landing velocity.

Our calculation results do align with our expectations for system performance. Since the performance of the rocket is heavily dependent on the body of the rocket, results will be covered in the aerostructures section.

2.4 Aerostructures

The rocket's aerostructure system concept is entirely defined by each primary component including the body tube, nose cone, and fins. Sub components such as the motor retainer, payload, avionics, and recovery systems. These components are clearly defined with respect to their roles in the design, balancing weight, internal, and external forces. Figure 12 details the rocket design sketched using OpenRocket. Sub components of the aerostructures system have also been sketched in CAD using SolidWorks.

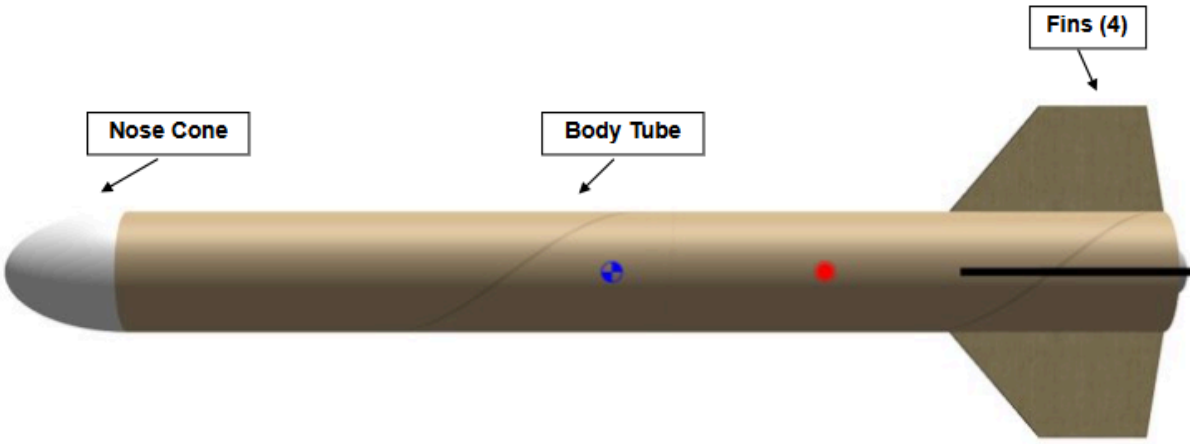


Fig 11. Overall Aerostructures Design

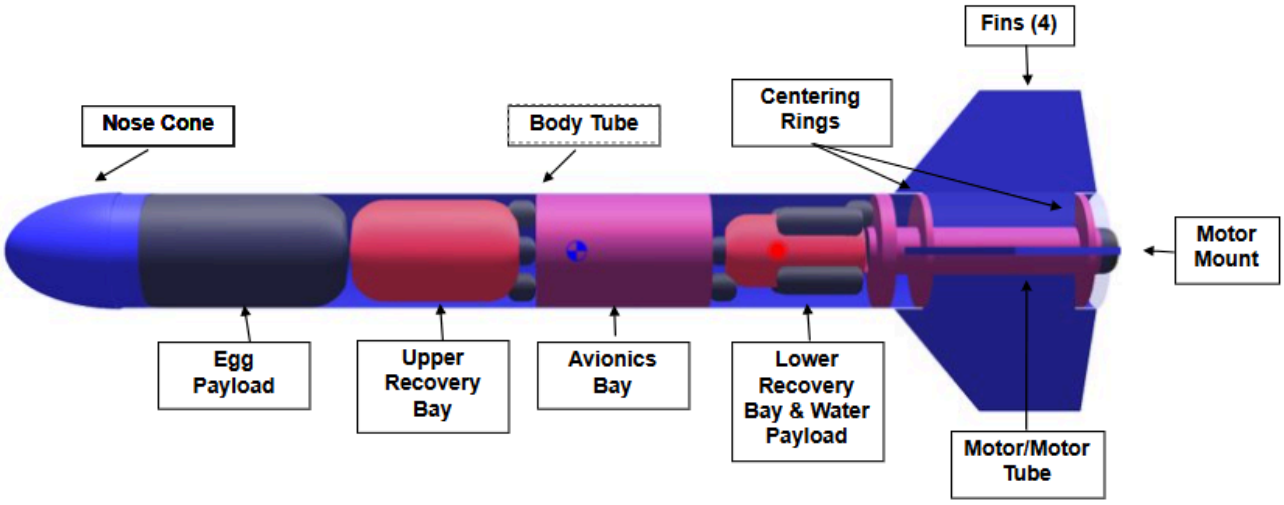


Fig 12. Detailed Aerostructures Design

Fundamental principles of flight mechanics were studied through OpenRocket simulation to understand flight parameters during specific points of launch, ejection, and landing.

Identifying areas of weakness in material selection and balancing the load within the rocket played a pivotal role in acquiring a successful simulation for prototyping. The various aerostructure components have also been modeled separately to prepare for maOpenRocket was used to evaluate and balance structural, aerodynamic, and flight parameters. Using this data, the team is able to understand how the rocket would perform under various flight conditions to anticipate how the rocket would react on competition day.

Applicable to the aerostructures system, the team can use principles of solid mechanics evaluating stress, strain, and buckling. Aerodynamics, vibrations, and materials science are also crucial topics for evaluation in this prototype. Using these methods of evaluation, the design was modeled with practical and analytical methods. Simulating flight data under various conditions to test the vehicles ability to meet competition requirements and understanding areas of potential failure. The team is able to create means of pre- and post-inspection evaluations of the vehicle to ensure it meets safety requirements for those critical phases of flight. Understanding the rocket stability, mass, thrust, and center of pressure are a few parameters vital in the design and analytical phase of the objective.

chining purposes. Fins, bulkheads, and centering rings can be found in Figure 13.

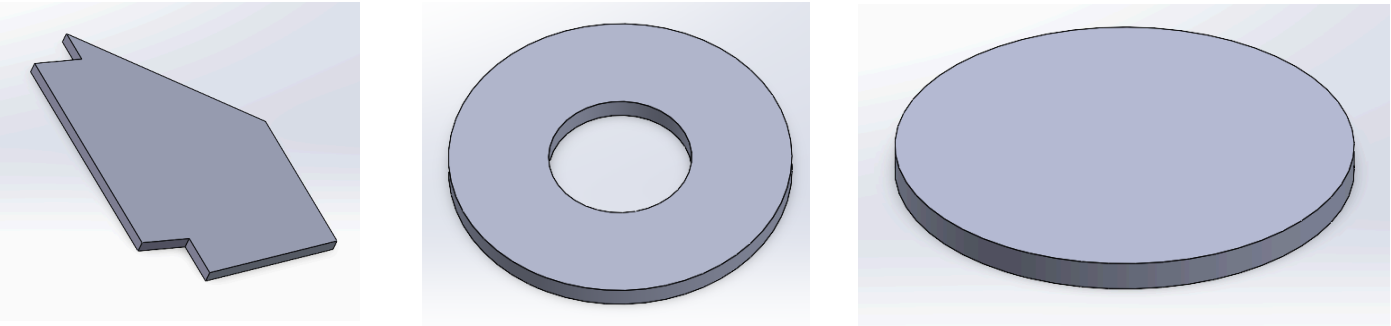


Fig 13. Fin, Centering Ring, & Bulkhead Design

Empirical data was gathered through simulation to obtain a reliable flight results that met the parameters of the competition. Through analyzing the average natural conditions, OpenRocket was able to create plots to represent the different aerodynamic conditions throughout flight. Altering the design also altered flight conditions and was critical to identify peak system performance. Figure 14 - Figure 20 shows the current design's flight capabilities to meet competition requirements.

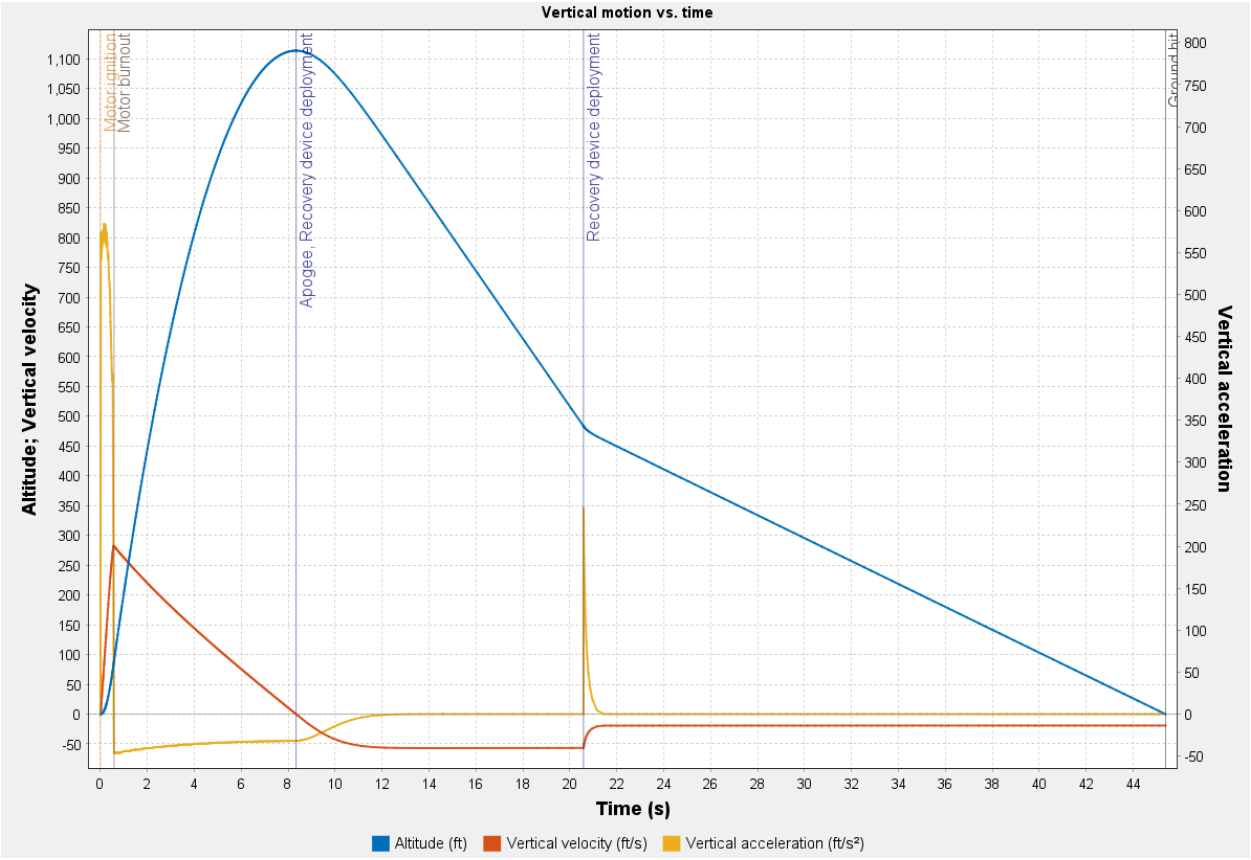


Fig 14. Vertical Motion vs. Time

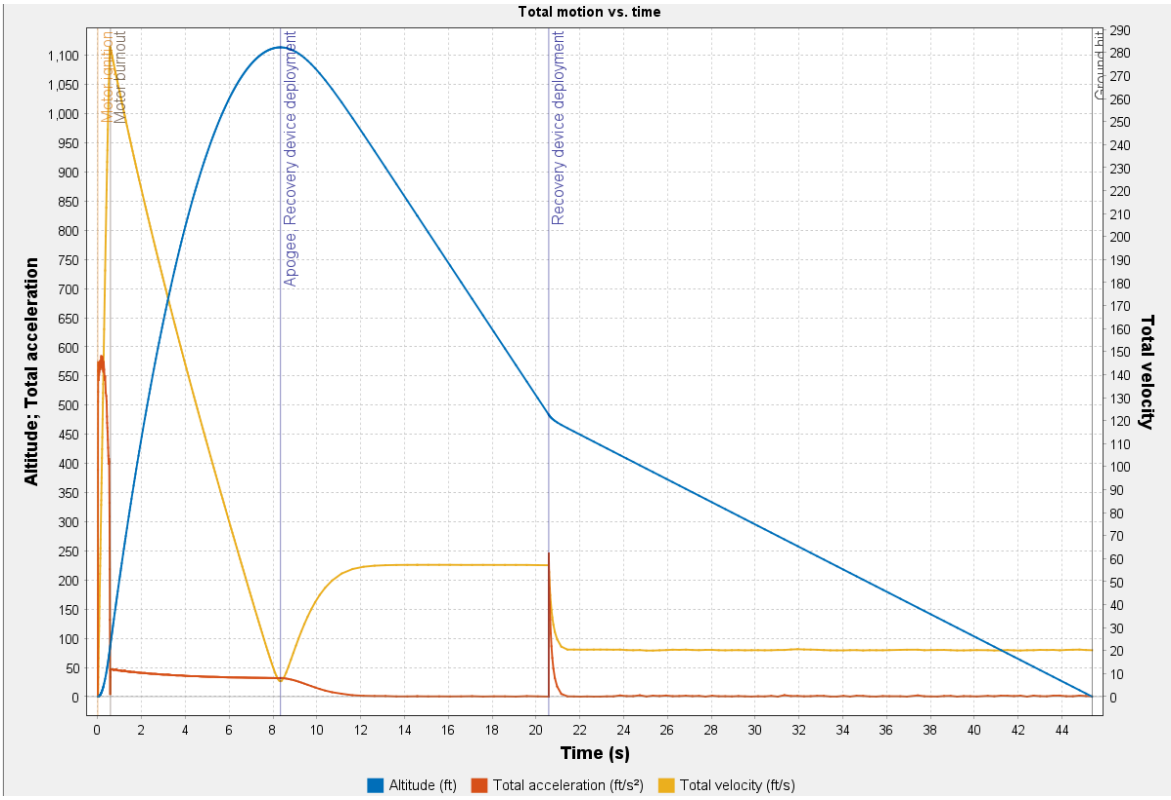


Fig 15. Total Motion vs. Time

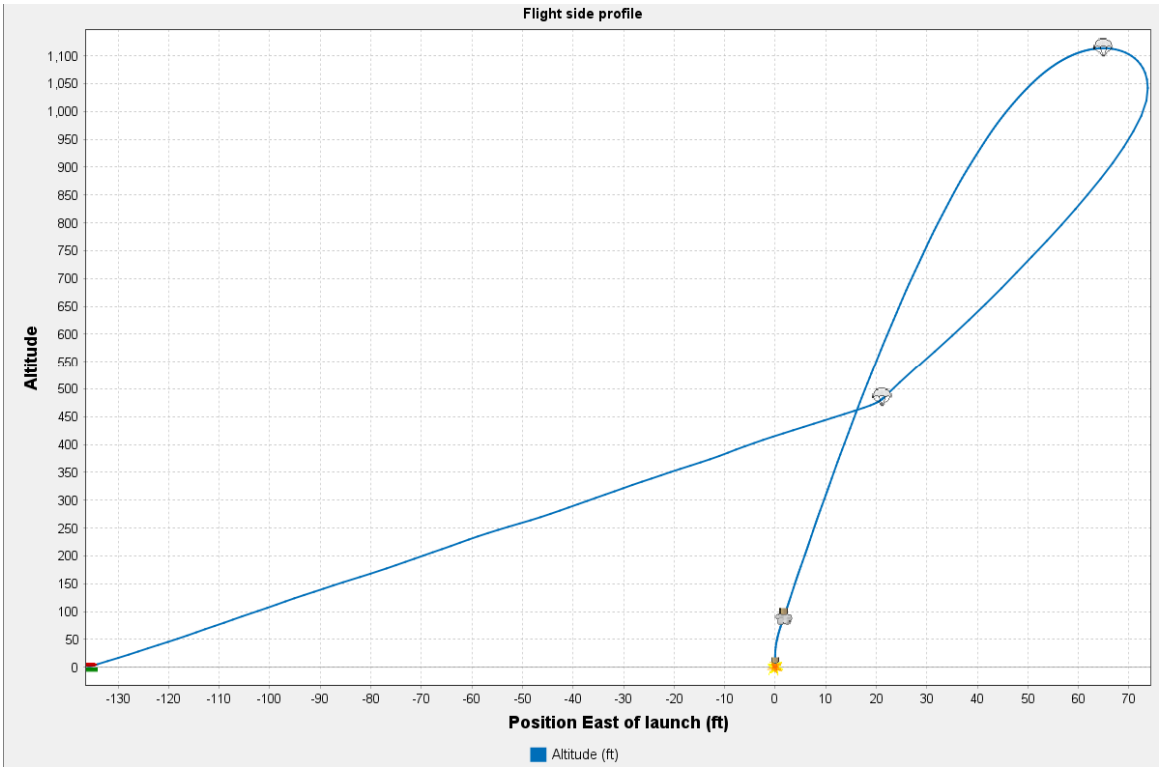


Fig 16. Flight Side Profile

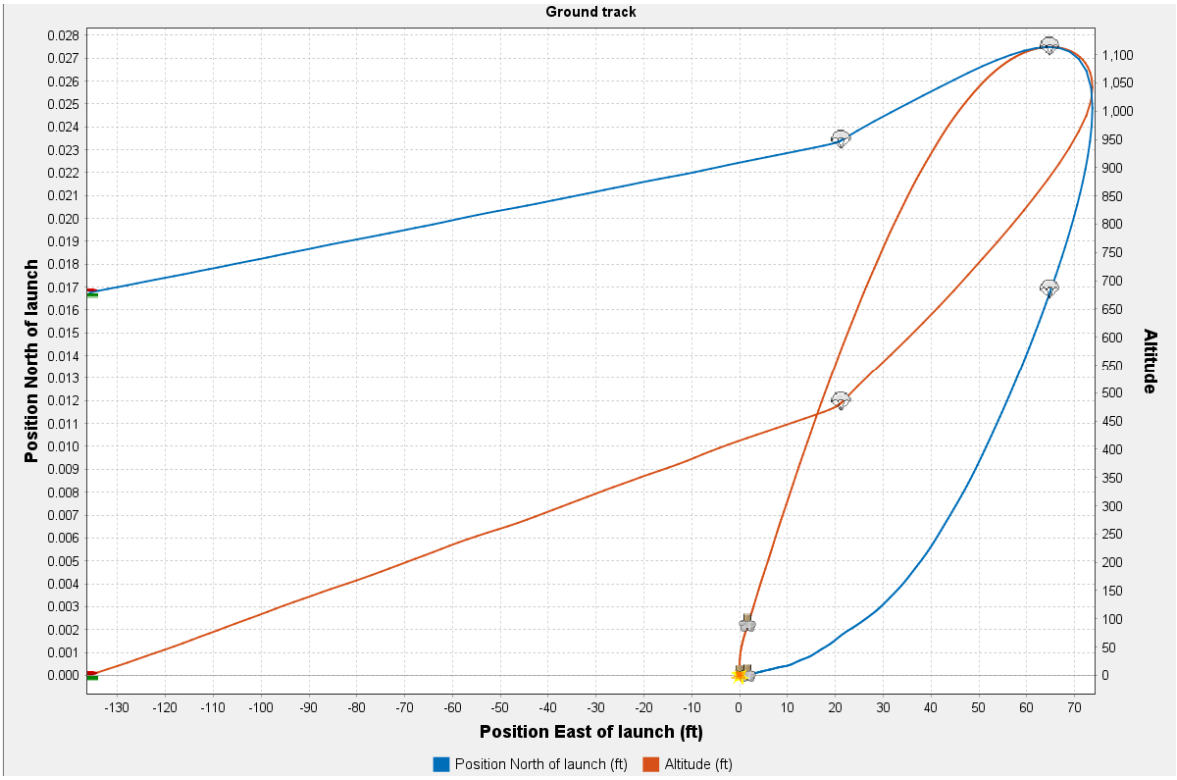


Fig 17. Ground Track

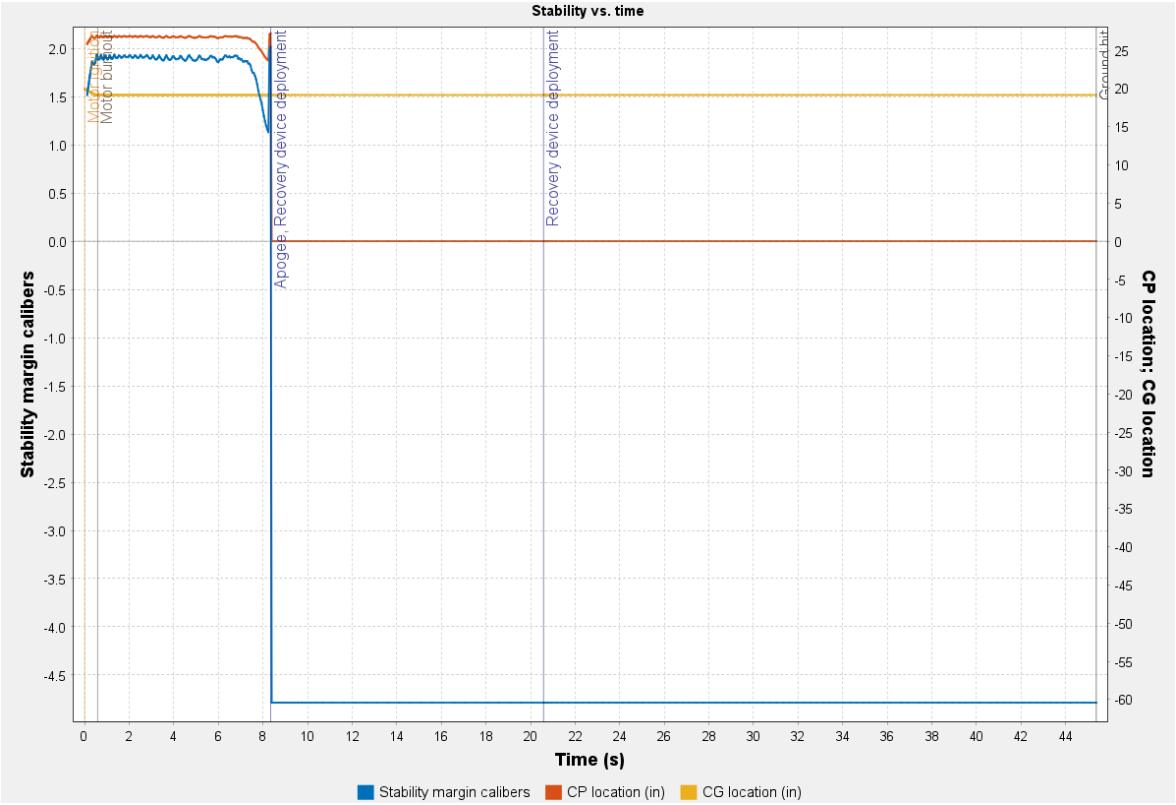


Fig 18. Stability vs. Time

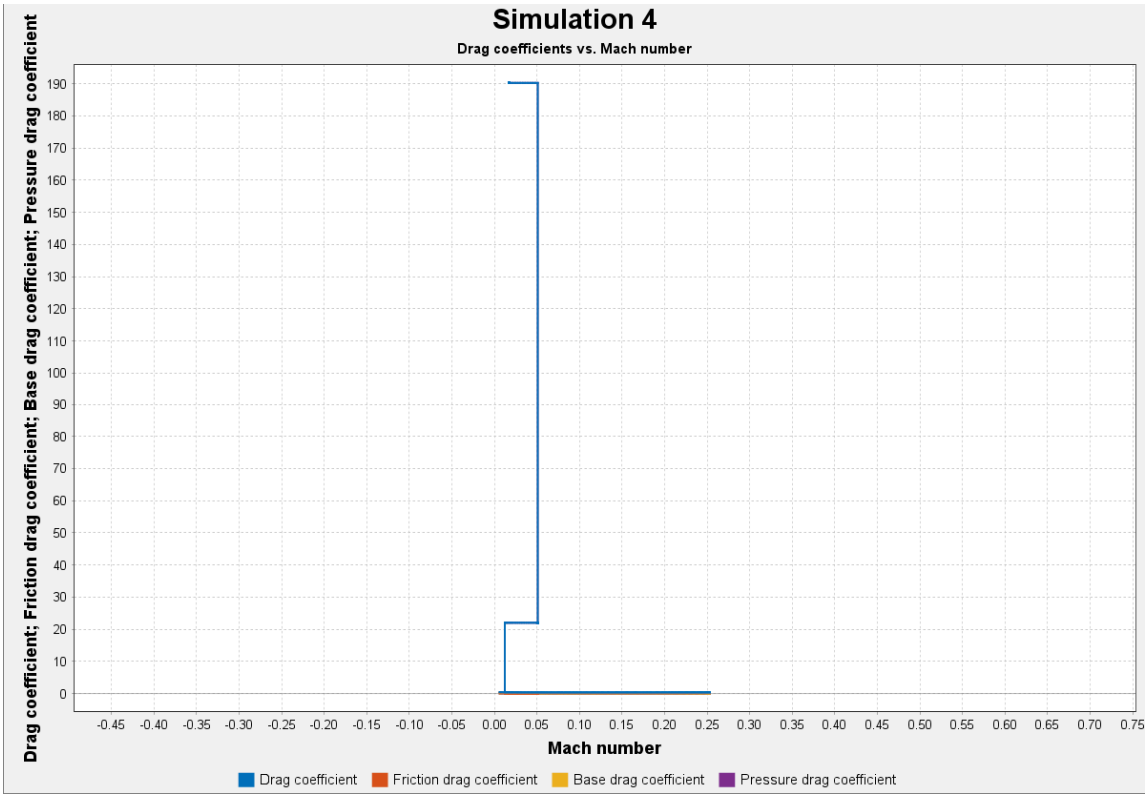


Fig 19. Aerodynamic Drag Coefficients vs. Mach Number

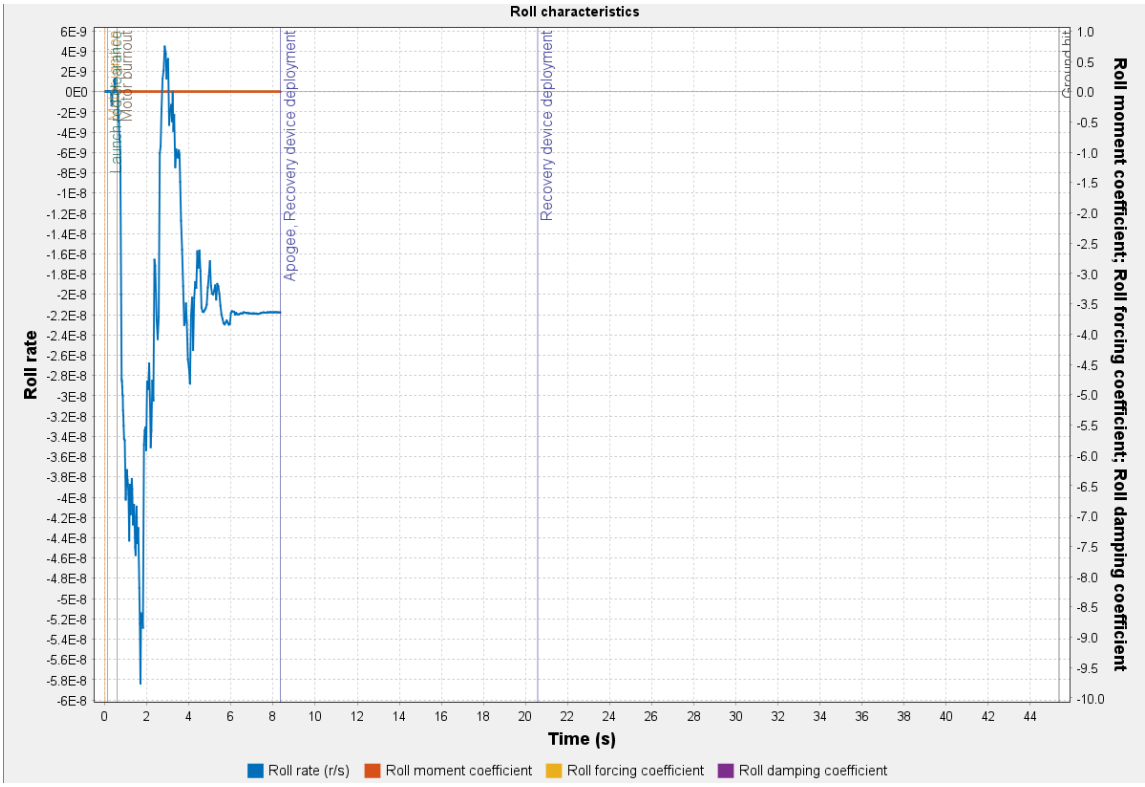


Fig 20. Roll Rate

2.5 Avionics

The avionics subsystem is considered complete and well defined because its architecture, component interfaces, operational requirements, and performance constraints have been explicitly specified and validated against mission objectives. The subsystem is responsible for flight event control, data logging, recovery actuation, tracking, and post-flight analysis. These functions require coordinated operation of multiple electronic components in a high-stress launch environment, making system definition and verification critical to mission success.

The current avionics configuration consists of a primary flight controller, redundant backup altimeter, dual-deployment recovery circuitry, battery power system, onboard camera, auditory recovery locator, and a GPS tracking device. This architecture was selected through structured trade studies balancing necessity, feasibility, affordability, and competition scoring potential. Redundancy was incorporated into flight-critical functions such as recovery deployment to reduce risk and improve reliability.

System visualization is supported through multiple engineering representations. CAD modeling defines component placement, mounting geometry, and structural clearances.

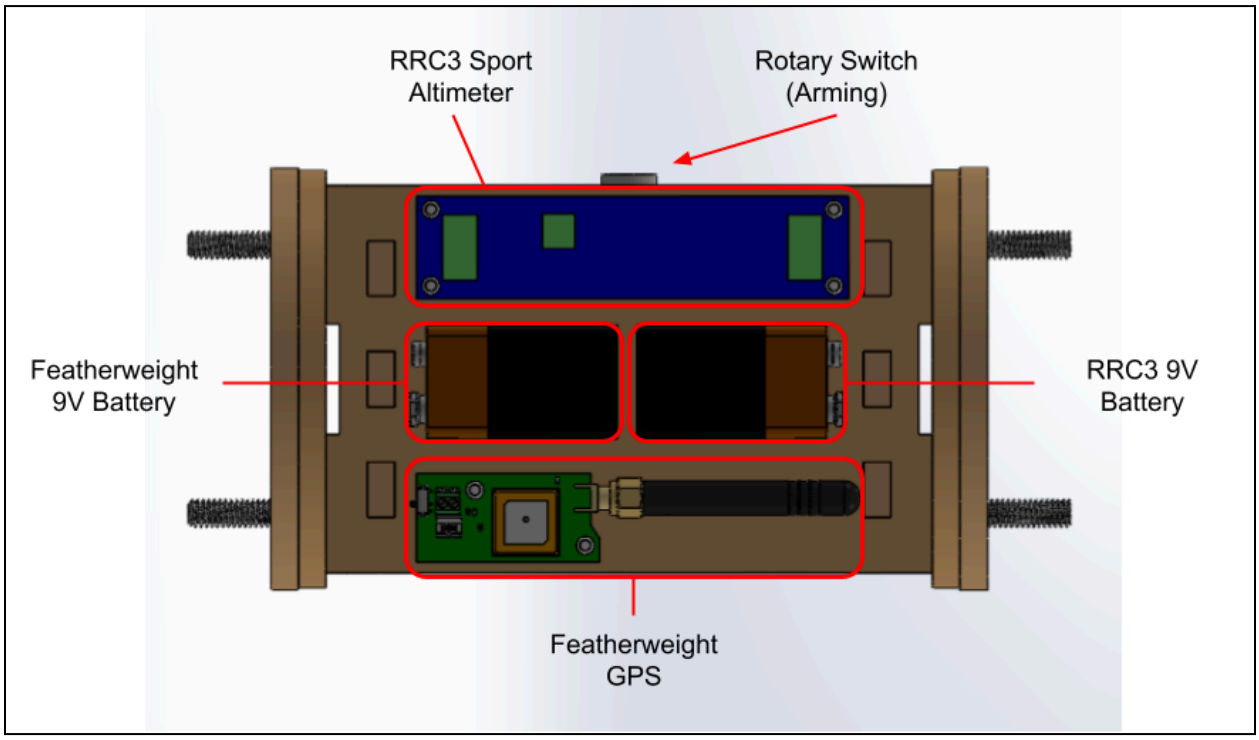


Fig. 21 Mounted Components in the Avionics Bay

Functional relationships between avionics and recovery compartments are also depicted in the determined configuration, as seen below.

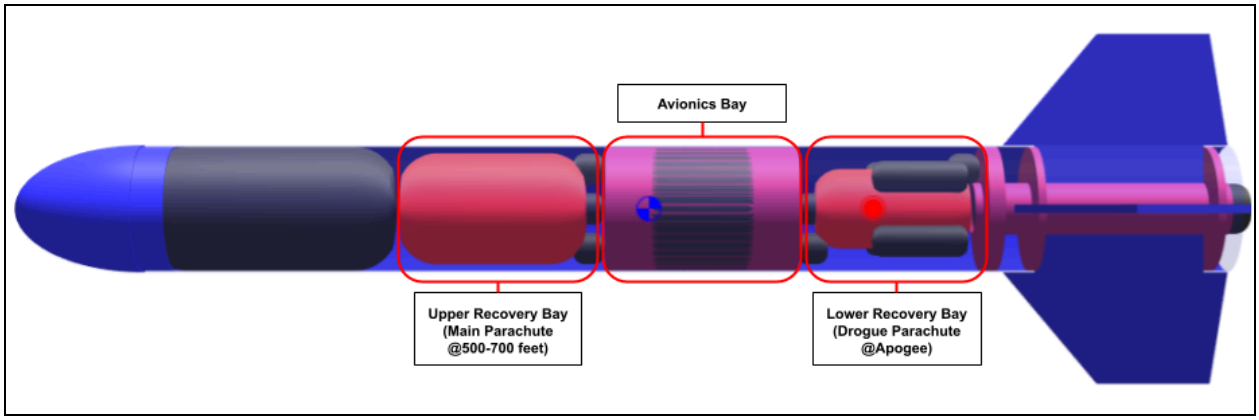


Fig 22. Avionics and Recovery Configuration

These models ensure proper integration with surrounding subsystems while verifying that venting, wiring access, and structural support requirements are satisfied.

The avionics design is governed by several engineering principles, including circuit analysis, flight event control, reliability engineering, flight mechanics, and environmental survivability. All of these principles were weaved throughout the process of developing this system, including in the inception of its engineering design specification and requirements. They were treated as constraints to ensure the system performed in accordance with its responsibilities. System performance constraints are formally defined in the table below.

Table 9. Avionics System Constraints

Constraint	Value	Requirement Referenced
Maximum Avionics Mass	20 ounces	AVIO - 6.1
Nominal Operating Voltage	9 Volts	AVIO - 6.4
Maximum Current Draw	500 milliAmps	AVIO - 6.5
Minimum Data Sampling Rate	10 Hertz	AVIO - 6.2
Minimum Data Recording Duration	5 minutes	AVIO - 6.3

These limits establish design boundaries ensuring compatibility with vehicle mass budgets and power availability.

Key design parameters considered during modeling include voltage stability, current demand, mass contribution, temperature tolerance, vibration loading, acceleration exposure, and humidity effects. Environmental boundary conditions are derived from predicted flight profiles

and include high acceleration loads, rapid pressure changes, temperature variation, and mechanical vibration.

The modeling and analytical approach is considered accurate because it uses validated simulation outputs, conservative design assumptions, verified component ratings, and redundancy for cross-checking critical measurements. Agreement between predicted performance and expected operating conditions indicates that the model sufficiently represents the physical system.

3.0 Prototyping and Testing

3.1 Egg Payload

Prototyping the egg payload capsule internals is necessary to mitigate the risks of the capsule failing during the competition. A partial prototype would not fulfill the needs for risk mitigation that have been outlined. Relying on analytical prototypes would not be as accurate. However, analytical prototypes are still necessary for downselecting and prioritizing possible configurations for the egg capsule design before manufacturing to lower the risk of wasting materials. A capsule of full scale can be made with cardboard, foam, a metal rod, and birch wood. The prototype can be tested by dropping it from a significant height to see if the eggs survive impact with the ground. The velocity on impact will be slightly above the team's desired impact velocity, and the height needed to reach this velocity can be calculated with the free-fall equation h= v^2/(2g). Plugging in a desired velocity of 20 ft/s gives the height 6.2ft. Passing the test if the eggs survive the drop, and failing if the eggs are damaged. Ideally, the parts purchased for the prototype can be reused for the competition, except for needing new foam. The goal of the test is to confirm that the foam can protect the eggs, identify the ideal arrangement of eggs, and identify the maximum number of eggs that can be held in the capsule safely. Development of the prototype will be deemed successful if the three goals are met.

After successful prototyping of the egg capsule, another test to reduce risk must be done for the integration of the capsule and the rocket. The test will be to see if the team is capable of loading the vehicle with eggs in under five minutes, then unloading in under five minutes. The test will reveal if there are potential issues like the capsule being stuck in the rocket, the capsule not opening, or if there is an issue with the team's process for egg transfer. This cannot be done with an analytical prototype, due to manufacturing tolerance factors and human factors. The remedy to issues that can arise from the test are easily reconciled by sanding the outside of the capsule, egg transfer workflow optimization, and practice.

3.2 Water Payload Subassembly

Creating a prototype is necessary to validate that the design is able to meet its requirements. Prototyping is also key for identifying issues early on, before committing to the final hardware. The primary objective for testing the water payload subassembly is ensuring that

the hardware is able to withstand forces comparable to what will be experienced during the competition flight. More specifically, ensuring the vessels will not have any unplanned leakage and that the supporting rings will not suffer structural damage.

For this subassembly, a full scale prototype will not be necessary, but also would not be impossible to produce as the overall system is relatively simple compared to the rest of the vehicle. A single, or multiple as needed, partial prototype will be used for physical testing in order to validate that the established requirements are satisfied. The prototype will undergo a few variations of stress testing, such as drop tests and flex tests to ensure structural integrity. As the main method of modelling the system is through SOLIDWORKS, incorporating tests on the analytical prototype would be able to provide a baseline for expectations on performance, however would not substitute for real world physical testing in this case.

Development efforts for prototyping and testing would be considered a success if it is clearly evident that the design concept is able to satisfy all of its requirements through reasonable testing procedures.

3.3 Recovery & Propulsion

Prototyping for recovery and propulsion isn't all that necessary. It is also difficult to test recovery and propulsion systems accurately without testing the rocket in its entirety. For recovery, it is possible to test that a parachute deploys, but difficult to test the deployment at the descent velocity our rocket will be travelling. The parachute is also tested by the manufacturer since we are not designing our own. For propulsion, it is not practical to prototype since our motor is already tested and validated by the manufacturer. Any testing that may be done on the recovery system would only be on the parachutes. Possible testing could be done by connecting a weight the same weight of our rocket to a parachute and doing a drop test from a high point, and seeing if the parachute slows down the weight to a viable speed before impact with the ground. The main issue with doing a test like this however, is that there are numerous other factors that can't be tested until we actually launch the rocket. If the parachute is able to slow down the weight enough then maybe it will slow down the rocket enough, but there is no way to test that to reach a definitive yes or no.

3.4 Aerostructures

Prototyping the aerostructures for this project is difficult as the end goal for this project will technically be the experimentation/prototype for the rocket body. Because aerostructures materials are expensive and encapsulate the entire body, it is difficult to test the capabilities of the rocket without launching it which will only be done once and that is on launch day. The primary object of prototyping this vehicle would be to verify structural integrity , confirm safety margins, assess tolerances and resistance to certain types of damage. Conveniently, multiple members of the team have launched rocket vehicles in similar capacities and similar materials to prove their effectiveness. The team can use that prior knowledge and apply to that of this project. Most prototyping for this project will include testing the capability and effectiveness of the payload capsules.

Partial prototypes such as the sub-components of the aerostructures system can be prototyped as their materials are less expensive and the manufacturing is less time consuming. With this, the team will be able to test the machining capability of the rocket fins, centering rings, and bulkheads by laser cutting. From there, testing can be done to verify the strength conditions of those components such as fin stiffness and constraints. Full scale prototyping would only be possible when the rocket is fully configured and built for launch which is when the team would be able to validate the joint strengths, mass distribution and load tests. In the meantime, the team can consult simulated variables done in OpenRocket to study the launch effects.

Testing the materials would involve testing different loads on the fin, bulkhead, and centering ring material. This information can also be found by finding the properties of the wood used for these components and even the cardboard used for the body tube.

Material	σ_{max} [MPa]	A_c [mm²]	Max Thrust [N]
Cardboard	11.0509	300.26	3319.28
Blue Phenolic	29.5440	516.66	15269.43
Fiberglass	175.1379	513.81	90018.35

Fig 23. Material Samples & Calculated Maximum Thrust [14]

Material	Type	TYS (psi)	UTS (psi)	UFS (psi)	E (ksi)	rho lb/cu.in.
Plywood	birch		6300	10750	1400	0.025
Garolite	G10		41000	75000	2700	0.065
	G11		43000	80000	3000	0.065
3D filament	PLA	6800	7300	12000	580	0.047
Aluminum alloy	1100	5000	13000		10300	0.098
	3003-H14		16000		10100	0.099
	5052-H22	22000	31000		10100	0.097
	5052-H34	25000	34000	48000	10100	0.097
	6061-T6	38000	43000	61000	9900	0.098
	2024-T3	48000	65000	90000	10500	0.101
	7075-T6	72000	80000	122000	10300	0.101

TYS = Tensile Yield Strength
UTS = Ultimate Tensile Strength
UFS = Ultimate Flexural Strength
E = Elastic Modulus
rho = mass density

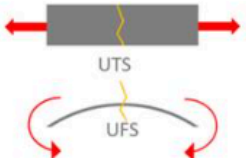


Fig 24. Material Samples & Strength Measurements [15]

Success of this project will be determined by the simulated parameters and further testing and validation of the payload sections, while also verifying the quality and capability of aerostructure sub-components. Safety margins have been assessed and will continue to be assessed for pre-flight checklists. The team will continue to evaluate potential failures and capability of materials as testing is validated and the overall build comes together.

3.5 Avionics

Prototyping is required because avionics performance cannot be verified through analysis alone. Electronic systems must be physically tested to ensure reliable electrical operation, correct deployment sequencing, sensor accuracy, and resistance to environmental stressors such as vibration and temperature variation. A complete full-scale prototype is not required for every validation step. Instead, testing will be conducted in stages.

Table 10. Avionics Testing Stages

Testing Stage	Name	Description
Stage 1	Bench Testing	Individual component functionality and electrical verification.
Stage 2	Subsystem Testing	Integrated avionics bay operation including wiring, mounting, and power routing.
Stage 3	Environmental Simulation Testing	Exposure to representative launch conditions such as vibration and load.

Partial prototypes are sufficient for validating sensors, timing logic, and recording capability, while full integration testing is reserved for final verification prior to launch readiness review. Testing methods include continuity checks, current draw measurement, ground-triggered deployment tests, vibration simulation, and timed data logging. These methods allow evaluation of both nominal and worst-case performance.

Prototype development will be considered successful if all flight events trigger at correct thresholds, data logs remain complete, electrical connections remain secure under vibration, deployment signals fire reliably, and the GPS unit acquires a signal within expected acquisition time. Meeting these criteria demonstrates readiness for system integration and flight operations.

4.0 Development Planning

4.1 Egg Payload

The egg payload system has been designed to be simple and effective. The internals of the capsule can be easily modified depending on the results of the prototype, while the exterior can be changed depending on the simulations of the nose-cone integrated design. The internal design has been proven to work in past CRASH competitions, varying depending on the type of foam selected. The objective of the design, besides meeting the goals of protecting the eggs and making eggs easily transferable, was to have a simple-to-manufacture design. The individual components that make up the system are the foam, the centering rod, the cardboard coupler, the plywood bulkheads, the ubolt, and the nosecone. Most parts will be purchased and modified as needed. The foam will be sanded down to fit in the coupler. The nose cone will be 3D printed. The bulkheads will be manufactured from plywood. The item with the longest lead time could be the nose cone due to printing failures, or the bulkheads due to machine shop wait times. All parts depend on the size of the coupler, so once it arrives, the foam can be sanded to fit inside, the nose cone can be printed, and the bulkheads can fit on the coupler. This system is the responsibility of Camila Martinez.

4.2 Water Payload Subassembly

The current design for the water payload subassembly has been chosen for its ease of application, and general low risk approach. The potential consequence of a failure of the water payload is disqualification from the competition, therefore low risk is much preferred. The primary objectives for the design are leak prevention and ease of loading, both requirements explicitly outlined in the competition ruleset.

The design is symmetric, and made of few individual components, totalling only six unique components: the vessels, caps, top and bottom support, nuts, and bolts. Only the supporting fixtures will require specific custom fabrication, while the rest of the hardware is available for purchase. As expected, the supports will then take the longest time to acquire. Critically, the design of the supports will be dependent on the specifications of the body tube for exact fitting. As the primary designer of this system, Christian Warson shall be responsible for all elements of the water payload subassembly.

4.3 Recovery & Propulsion

The current design for recovery and propulsion systems has been chosen for its simplicity and proven reliability. Since we are operating on a tight budget, we want to make sure that the majority of the money goes to developing other systems in order to enhance survivability of the payload, and keeping the recovery and propulsion systems simple can help with that. A reliable

system also increases our chances for success. The objectives are to successfully launch and recover the rocket in a flightworthy condition.

The recovery system is made up of 8 components: main parachute, drogue parachute, two shock cords, zipper protector, thermal parachute cover, and two quick links. The propulsion system is made up of just 1 component being the motor. All components besides the zipper protector will be purchased, and the zipper protector will be fabricated through the use of a 3D printer. The only long lead time component is the motor. This is due to it being made of explosive material, which requires certain transportation rules such as only being allowed to ship by ground, needing safer handling as compared to other package types, and only being transported by certain companies.

The critical path for the development of these systems are to first have the fabrication of the body completed, so that the parachutes, shock cords, and motor can be installed. Then the electronics bay needs to be completed so the shock cords can attach it to the rocket. The responsibility of both the recovery and propulsion systems falls on John Bray.

4.4 Aerostructures

The current rocket design being pursued is an aerostructure configuration to achieve reliable, stable, and safe flight. Safely withstanding thrust, aerodynamic forces, and other loads is required to meet competition requirements. The main goal is to ensure structural integrity while properly housing all payload and avionics compartments. Minimizing risks to the overall structure is crucial to the success of flight as degradation of aerostructure components could lead to catastrophe.

Combining all the necessary aerostructure components of the body tube, nose cone, fins, centering rings, bulkheads, couplers, and motor retainer make up the entire system which will mainly be purchased as pure material and fabricated to their design specification. The couplers and body tubes will be purchased with machining required to the body tube to break it into sections for recovery separation and fin slots to maintain proper flight. Fins, centering rings, and bulkheads will all be laser cut out of wood planks to meet specification. Nose cone and motor retainer will both be precisely printed using a 3D printer. The bulk of lead time for these components comes from the material acquisition, but machining these components can be done in as little as one day depending on machine availability.

For critical development effort, Logan Hammer will take lead of machining and overseeing the completion of aerostructure components and will schedule accordingly with UCF machine shops and labs to have these pieces properly fabricated. Timing is critical at this stage as machine shop availability can be limited, but the manufacturing itself takes very little time.

4.5 Avionics

The avionics subsystem is pursued because it is essential for safe and successful mission execution. It provides flight control authority, recovery actuation, and data acquisition, all of which are required for vehicle safety, competition compliance, and performance evaluation. Without a reliable avionics system, the vehicle would fail operational requirements and risk disqualification.

The design objectives for the system are based around a number of broad goals made to ensure the avionics effectively performs the tasks assigned to it in the final operation of the vehicle. In the final vehicle configuration, the avionics must perform reliable flight event control, accurate data recording for post-flight analysis, redundant recovery deployment capability, establish a safe arming and ground handling procedure, and have effective post-flight location capability.

The avionics system consists of multiple components, including a flight computer, backup altimeter, battery, switches, wiring harnesses, pyrotechnic outputs, camera, GPS tracker, recovery locator, and structural avionics bay housing. Each component fulfills a defined functional objective within the overall architecture.

Commercial off-the-shelf electronics are used wherever possible to reduce development risk and ensure reliability. Fabrication is limited to mechanical integration components such as mounting plates, structural housings, and harness routing hardware. This strategy minimizes complexity while maintaining system survivability.

Long-lead components represent a primary schedule risk. Electronics, motors, recovery devices, and specialty hardware can experience procurement delays that may interrupt testing or assembly. To mitigate this risk, components are ordered early and alternate suppliers are identified when possible.

In order to establish an effective critical development path, the system will follow a process of bench testing, bay assembly, integrated testing, vehicle integration, and finally be deemed ready for launch readiness. The responsibility for the development and integration of this path falls to team member Dylan Moga.

References

[1] U.S. Plastic Corp., “Knowledge Base: HDPE Properties,” URL: https://www.usplastic.com/knowledgebase/article.aspx?contentkey=1001

[2] Pandey, J. et al., “High-Density Polyethylene: Properties and Applications,” *Polymers*, Vol. 14, No. 12, 2022, URL: https://www.mdpi.com/2073-4360/14/12/2415

[3] Laird Plastics, “HDPE Guide—Properties, Uses, Applications (2025 Update),” URL: https://lairdplastics.com/resources/hdpe-guide-properties-uses-applications-2025-update/

[4] Precision Tork, “Is Your Product’s Cap Tight Enough?” URL: https://www.precisiontork.com/is-your-products-cap-tight-enough/

[5] Bolt Science, “Vibration Loosening in Bolted Joints,” URL: https://www.boltscience.com/pages/vibloose.htm

[6] MIT OpenCourseWare, “Cellular Solids: Structure, Properties and Applications, Lecture 16,” URL: https://ocw.mit.edu/courses/3-054-cellular-solids-structure-properties-and-applications-spring-2015/d7902e18aed019e6af0651e84a55e703_MIT3_054S15_L16_enab_trans.pdf

[7] Foam Factory, Inc., “Polyethylene Foam Sheet,” URL: https://www.foambymail.com/polyethylene-foam-sheet.html

[8] Foam Factory, Inc., “Cross-Linked Polyethylene Sheet,” URL: https://www.foambymail.com/cross-linked-polyethylene-sheet.html

[9] ADMET, Inc., “Egg Compression Strength Test,” *ADMET Blog*, URL: https://www.admet.com/blog/egg-compression-strength-test-2/

[10] NASA Glenn Research Center, “Newton’s Laws of Motion,” URL: https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/newtons-laws-of-motion/

[11] Engineering Toolbox, “Water – Density and Specific Weight,” URL: https://www.engineeringtoolbox.com/water-density-specific-weight-d_595.html

[12] Engineering Toolbox, “Hydrostatic Pressure from Water,” URL: https://www.engineeringtoolbox.com/hydrostatic-pressure-water-d_1632.html

[13] W. Nakka, “Recovery System,” *Nakka Rocketry*, URL: https://www.nakka-rocketry.net/RD_recovery.htm

[14] Toft, T., “Compression Tests of Tubing Used in Rocketry,” Apogee Components. URL: https://www.apogeerockets.com/education/downloads/Newsletter510.pdf

[15] Nakka, R., “Introduction to Rocket Design - Fins,” Richard Nakka Experimental Rocketry. URL: https://www.nakka-rocketry.net/RD_fin.htm#:~:text=However%2C%20even%20if%20a%20fin,four%20fins%20due%20to%20flutter

Appendices

Appendix A: List of Figures

Figure	Description	Page
Fig 1	Slide from MIT Structures lesson Cellular Solids: Structure, Properties and Applications [6]	20
Fig 2	Slide from MIT Structures lesson Cellular Solids: Structure, Properties and Applications [6]	21
Fig 3	Slide from MIT Structures lesson Cellular Solids: Structure, Properties and Applications [6]	22
Fig 4	Foam Impact Design Calculator	23
Fig 5	3.5 in PE 2.2	24
Fig 6	2in PE 2.2 EGG 3	25
Fig 7	2in PE 2.2 EGG 2	25
Fig 8	Nose cone integrated payload	27
Fig 9	Water Payload Subassembly Models	28
Fig 10	Model of recovery system in a dual deployment rocket [13]	29
Fig 11	Overall Aerostructures Design	31

Fig 12	Detailed Aerostructures Design	31
Fig 13	Fin, Centering Ring, & Bulkhead Design	32
Fig 14	Vertical Motion vs. Time	33
Fig 15	Total Motion vs. Time	34
Fig 16	Flight Side Profile	34
Fig 17	Ground Track	35
Fig 18	Stability vs. Time	35
Fig 19	Aerodynamic Drag Coefficients vs. Mach Number	36
Fig 20	Roll Rate	36
Fig 21	Mounted Components in the Avionics Bay	37
Fig 22	Avionics and Recovery Configuration	38
Fig 23	Material Samples & Calculated Maximum Thrust [14]	42
Fig 24	Material Samples & Strength Measurements [15]	42

Appendix C: List of Tables

Table	**Description**	**Page**
Table 1	Egg Payload Risk Assessment	3-4
Table 2	Water Payload Risk Assessment	5
Table 3	Recovery & Propulsion Risk Assessment	8-10
Table 4	Aerostructures Risk Assessment	11-16

Table 5	Avionics Risk Assessment	17-19
Table 6	Comparison of PE and XPLE Foams	22
Table 7	Egg Capsule variation measurements	25-26
Table 8	Egg Capsule Decision Matrix	26
Table 9	Avionics System Constraints	38
Table 10	Avionics Testing Stages	43

Appendix D: Terms and Abbreviations

Abbreviation	**Term**
COTS	Commercial Off-the-Shelf
CRASH	Controlled Recovery and Safe Handling
HDPE	High-Density Polyethylene
TPU	Thermoplastic PolyUrethane